

Exact Gaussian range laws for structured sketches

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Abstract

When does a structured sketch produce exactly the same approximation as a dense Gaussian sketch *in distribution*? For a fixed input support and independent columns, we prove a necessary and sufficient condition: each projected column is nonzero and its line is uniform on projective space. This identifies exact Gaussian laws for rangefinders, PSD Nyström, and powered or Krylov ranges. Conditional scalar Gaussianity supplies explicit Khatri–Rao and tensor-train examples. On these supports, the classical Nyström coefficient $r/(k-r-1)$ holds without a tensor-dependent penalty. The restriction is substantial: product probes require support dimension at most $1 + \sum_j (n_j - 1)$, and the conditional product class reduces to an $n_1 \times n_1$ matrix. We exploit that reduction to transfer Gaussian generalized Nyström and trace estimators to legal product queries. Counterexamples show why the full support hypothesis, left-side calibration, and the distinction between range and embedding laws are essential.

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1 Introduction

Let $A \in \mathbb{R}^{N_L \times N}$ be a fixed matrix and $\Omega \in \mathbb{R}^{N \times k}$ a random test matrix. A rangefinder forms $Y = A\Omega$ and approximates A by $P_Y A$, where P_Y is the orthogonal projector onto $\text{range } Y$. Multiplying one column of Y by a nonzero scalar does not change P_Y . Neither does multiplying a column of Ω by a nonzero scalar. This elementary fact has a sharp consequence for structured sketches: whatever random column scales a structured probe carries, they are invisible to every output that depends only on the spanned subspace.

The question we answer is: *exactly when does a structured sketch give the same output law as a dense Gaussian sketch, for range-determined outputs?* The answer (Theorem A) is a statement about one column: its projection onto the relevant support must have a uniformly distributed line. The answer is exact, not asymptotic.¹

The practical interest comes from tensor-structured probes such as Khatri–Rao products $g \otimes h$ and tensor trains. These have cheap matrix–vector products in tensor formats, but their projected coordinates share random scales, and their sketch Gram matrices are not Wishart. On the class identified here, those scales cancel from the range, and the sharp Gaussian constants hold

¹The phrase “Gaussian equivalence” is also used for asymptotic universality of random feature models. The statements here are exact and finite-dimensional.

verbatim. For example, common-Schmidt Khatri–Rao Nyström has expected trace error at most $(1 + r/(k - r - 1))$ times optimal (Corollary 7.5). This is the classical Gaussian coefficient. Its appearance here follows from equality of output laws, rather than from an improved embedding estimate.

We are equally explicit about what the class is. A dimension count shows that a product probe can meet the criterion only on supports of dimension at most $\sum_j (n_j - 1) + 1$ (Theorem B). For the conditionally Gaussian product class the rank is at most n_1 , and the whole problem reduces exactly to an $n_1 \times n_1$ matrix reachable by one product query per matrix–vector product (Theorem C). The class-specific results should therefore be read as exact identifications and as clean product-query transfers of Gaussian algorithms. Their algorithmic content is largest when n_1 is large compared with the target rank or accuracy budget.

1.1 Prior work and the contribution of this paper

Gaussian rangefinding and its pseudoinverse analysis are classical [6]. The PSD approximation and truncation bounds used below come from Tropp, Yurtsever, Udell and Cevher [17]; the corresponding two-sketch reconstruction is developed in [18, 12]. Hutch++ and Nyström++ already establish the $O(\varepsilon^{-1})$ trace-estimation principle [7, 13]. We include proofs of the Gaussian estimates needed here to isolate exactly what is transferred to product queries.

For structured sketches, Bujanović, Grubišić, Kressner and Lam [1], Saibaba, Verma and Ballard [16], and Beretta and Musco [2] study Khatri–Rao embeddings or approximation guarantees on general fixed subspaces. Camaño, Epperly, Meyer and Tropp [4] organize structured approximation around oblivious subspace injections. Those results address broad input classes. The condition studied here is stronger and holds on a restricted full support: it identifies the entire finite-sample output law, including its tails, rather than only an error guarantee.

The contribution is the use of projective uniformity as a complete criterion for these output laws, together with concrete support certificates, a rank obstruction, and an exact product-query reduction. The rotation invariance, inverse-Wishart moments, Hurwitz identities [9], and Gaussian algorithms underlying the argument are established facts. The new class-specific consequences are one-response calibration and the calibrated or frozen-environment implementations. The numerical evidence supports these identities; it does not establish priority or a general-input improvement.

1.2 Setting and notation

Throughout, $\|\cdot\|$ is the Euclidean norm on vectors, $\|\cdot\|_F$ the Frobenius norm, and $M^\dagger$ the Moore–Penrose pseudoinverse. For a PSD matrix A with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ we write $\tau_r(A) = \sum_{j>r} \lambda_j(A)$; for a rectangular matrix $\tau_r^{(2)}(A) = \sum_{j>r} \sigma_j(A)^2$. The Grassmannian of j -dimensional subspaces of $\mathbb{R}^R$ is $\text{Gr}(j, R)$, and $\mathbb{RP}^{R-1} = \text{Gr}(1, R)$. We write $\mathcal{G}_R = \bigsqcup_{j=0}^R \text{Gr}(j, R)$ for the space of all subspaces, with its Borel structure. For a nonzero vector x , $[x]$ is the line it spans.

A *probe* is a random vector $\omega \in \mathbb{R}^N$. A *sketch* of size k is $\Omega = [\omega_1 \ \dots \ \omega_k]$. Unless explicitly stated otherwise, the columns are independent copies of ω . When a probe is generated with auxiliary randomness (an *environment*), the pairs (probe, environment) are independent across columns; the one exception is the frozen-environment designs of Sections 8 and 9, where it is stated explicitly.

The *relevant support* $S \subseteq \mathbb{R}^N$ of a fixed input is the row space of A for rectangular algorithms, and the range of A for PSD algorithms. We assume $R = \dim S \geq 1$ and fix an isometry $V \in \mathbb{R}^{N \times R}$ onto S , and compare with a dense Gaussian sketch $\Gamma \in \mathbb{R}^{N \times k}$ with iid $N(0, 1)$ entries. A *product probe* on $\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_q}$ is $\omega = x_1 \otimes \cdots \otimes x_q$ with random factors $x_j \in \mathbb{R}^{n_j}$; a *Gaussian Khatri–Rao probe* has independent $N(0, I_{n_j})$ factors. We put

$$\delta = \sum_{j=1}^q (n_j - 1),$$

the projective dimension of the Segre variety of rank-one tensors. All statements concern exact arithmetic on a fixed input. The zero input is handled directly and needs no projective-space criterion.

2 Main results

Definition 2.1 (Range-determined output). An output $\mathcal{O}(\Omega)$ of a sketching algorithm on a fixed input is *range-determined on S* if $\mathcal{O}(\Omega) = F(\text{range}(V^\top \Omega), \Xi)$ for a measurable map F and an auxiliary random element Ξ independent of Ω .

Section 3 shows that rangefinder projectors, powered and block Krylov ranges, PSD Nyström outputs (including the fixed-rank truncation), and generalized Nyström with an independent Gaussian left sketch are all range-determined; Gram matrices and embedding distortions are not.

Theorem A (Exact range-law criterion). *Let the columns of Ω be independent copies of a probe ω . The following are equivalent.*

- (i) *For every $k \geq 1$, $\text{range}(V^\top \Omega)$ has the same law as $\text{range}(V^\top \Gamma)$; hence every range-determined output has its dense-Gaussian law for every k .*
- (ii) *The rangefinder with $k = 1$ has its Gaussian law: $\text{range}(A\omega)$ and $\text{range}(A\gamma)$ have the same law for one (equivalently every) A with row space S .*
- (iii) *$V^\top \omega \neq 0$ almost surely, and $[V^\top \omega]$ is uniformly distributed on $\mathbb{RP}^{R-1}$.*

A checkable sufficient condition is conditional scalar Gaussianity: $V^\top \omega \mid \mathcal{F} \sim N(0, wI_R)$ with $w > 0$ almost surely and $\mathcal{F}$ -measurable (Theorem 4.2).

When (iii) holds we say that the probe has the *Gaussian range law on S* ; for brevity we also say that S satisfies Theorem A.

Theorem B (Rank ceiling). *Let $\omega = x_1 \otimes \cdots \otimes x_q$ be a product probe with arbitrary, possibly dependent, factor laws, and $\delta = \sum_j (n_j - 1)$.*

- (i) *If the conditions of Theorem A hold, in particular under conditional scalar Gaussianity, then $R \leq \delta + 1$. This is attained.*
- (ii) *If for one sketch size $k < R$ the law of $\text{range}(V^\top \Omega)$ equals the Gaussian one, then $R \leq k + \delta$, whatever the joint law of the columns.*
- (iii) *If the criterion holds conditionally on $x_2, \dots, x_q$ with x_1 Gaussian (the conditionally Gaussian product class), then $R \leq n_1$.*

Thus the nontrivial window for the class-specific approximation theorems is $r + 1 < k < R \leq \delta + 1$. With base dimension two ($n = 2^q$), the ceiling is $\log_2 n + 1$. For Gaussian tensor-train probes the corresponding bound is the dimension of the tensor-train variety, which grows with the bond dimension; growing rank is not excluded there, and we do not claim it (Remark 6.3).

Theorem C (The class is an $n_1 \times n_1$ matrix in disguise). *Let $A \succeq 0$ have range S and suppose the Gaussian product probe $\omega = g \otimes h$, $h = h_2 \otimes \cdots \otimes h_q$, satisfies $V^\top \omega \mid h \sim N(0, w(h)I_R)$ with $w > 0$ almost surely. Put $L_h x = x \otimes h$. Then for every product environment h with $w(h) > 0$ (almost every h):*

- (a) $w(h) = \|L_h^\top v\|^2 / \|v\|^2$ for every nonzero $v \in S$;
- (b) $Av = A(L_h L_h^\top v / w(h))$ for every $v \in S$, and $L_h L_h^\top v / w(h) = (L_h^\top v / w(h)) \otimes h$ is one legal product query;
- (c) $B = L_h^\top A L_h / w(h) \in \mathbb{R}^{n_1 \times n_1}$ has the same positive eigenvalues as A , with multiplicities, and each product Bx costs one product query;
- (d) the n_1 product queries $e_i \otimes h$ recover A exactly, so $R \leq n_1$ and $\text{tr } A = \text{tr } B$;
- (e) powered and Krylov ranges through $A^{t+1}\Omega$ cost $k(1+t)$ product queries; for rectangular inputs whose two supports are in the class, t cycles of $AA^\top$ cost $k(1+2t)$.

Theorem D (Generalized Nyström with structured sketches). *Let the column space of A be in the conditionally Gaussian product class for the left probes, let $Y = A\Omega$ have rank k , and let $p > k + 1$.*

- (i) Calibrated. *Rescaling the p stored left measurements by weights read from one response gives an estimator with $\mathbb{E}[\|A - \hat{A}\|_F^2 \mid Q] = (1 + \frac{k}{p-k-1}) \|(I - QQ^\top)A\|_F^2$, where Q is a basis of $\text{range}(A\Omega)$.*
- (ii) Frozen environment. *If all left probes share one environment, the unmodified formula $Y(\Psi^\top Y)^\dagger \Psi^\top A$ satisfies the same identity, with no calibration.*

If the row space also satisfies Theorem A, both outputs have exactly the two-Gaussian law, and $k = 2r + 2$, $p = 2k + 2$ give $\mathbb{E}\|A - \hat{A}\|_F^2 \leq 4\tau_r^{(2)}(A)$ with $6r + 8$ product queries.

Theorem E (Calibrated product-query trace estimation). *For $A \succeq 0$ in the conditionally Gaussian product class, a nonadaptive unbiased estimator using $\min\{2\lceil \sqrt{6\kappa_G}/\varepsilon \rceil, n_1\}$ real product queries satisfies $\mathbb{P}\{|\hat{T} - \text{tr } A| > \varepsilon \text{tr } A\} \leq 1/3$, where $\kappa_G = e^{2-\gamma}/2 = 2.0743\dots$ and γ is Euler's constant. The estimator is exact almost surely once its approximation sketch has at least rank A columns.*

The boundary results (Section 11) are exact: a Gaussian range law is not a Gaussian embedding law; a common-Schmidt hypothesis on the leading eigenvectors alone changes the Nyström error by an explicit factor $\Phi(y) > 1$; and left rescaling of two structured sides changes generalized Nyström.

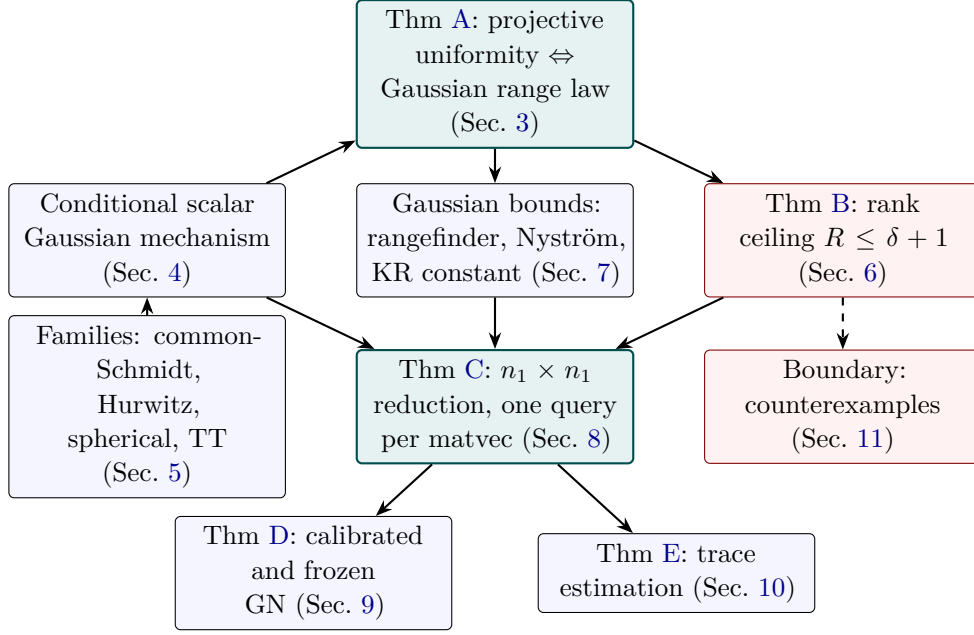


Figure 1: Roadmap. Teal boxes are the two structural theorems; red boxes mark the limits of the class.

3 Range-determined outputs and the exact criterion

3.1 Which outputs are range-determined

The basic identity is that a fixed linear map B with $BV_\perp = 0$, where $[V \ V_\perp]$ is orthogonal, satisfies $B = BVV^\top$, hence

$$\text{range}(B\Omega) = BV \text{ range}(V^\top \Omega). \quad (3.1)$$

Lemma 3.1 (Catalogue of range-determined outputs). *The following outputs are range-determined on S in the sense of Definition 2.1.*

- (a) *The rangefinder projector $P_{A\Omega}$, where S is the row space of A .*
- (b) *The joint ranges of $Y_j = (AA^\top)^j A\Omega$, $j = 0, \dots, s$, and of their concatenation $[Y_0 \ \dots \ Y_s]$; for $A \succeq 0$ with range S , the ranges of $f_j(A)\Omega$ and their concatenation whenever $f_j(0) = 0$.*
- (c) *The PSD Nyström approximation $C = A\Omega(\Omega^\top A\Omega)^\dagger \Omega^\top A$ for $A \succeq 0$ with range S , and its best rank- r truncation C_r (with a fixed measurable tie-breaking rule).*
- (d) *Generalized Nyström $\hat{A}_{\text{GN}} = A\Omega(\Psi^\top A\Omega)^\dagger \Psi^\top A$ with a left sketch Ψ independent of Ω and injective on $\text{range}(A\Omega)$ almost surely (e.g. dense Gaussian with $p \geq k$ columns).*

Proof. Rangefinder. A annihilates $S^\perp$, so (3.1) applies, and AV is injective.

Powered ranges. Each $(AA^\top)^j A$ and each $f_j(A)$ with $f_j(0) = 0$ annihilates $S^\perp$. The range of the concatenation is $\sum_j B_j V \text{ range}(V^\top \Omega)$.

PSD Nyström. With $M = A^{1/2}\Omega$, the identity $M(M^\top M)^\dagger M^\top = P_{\text{range } M}$ gives

$$C = A^{1/2} P A^{1/2}, \quad P = P_{\text{range}(A^{1/2}\Omega)}, \quad (3.2)$$

and $\text{range}(A^{1/2}\Omega) = A^{1/2}V \text{ range}(V^\top \Omega)$. The truncation is a measurable function of C .

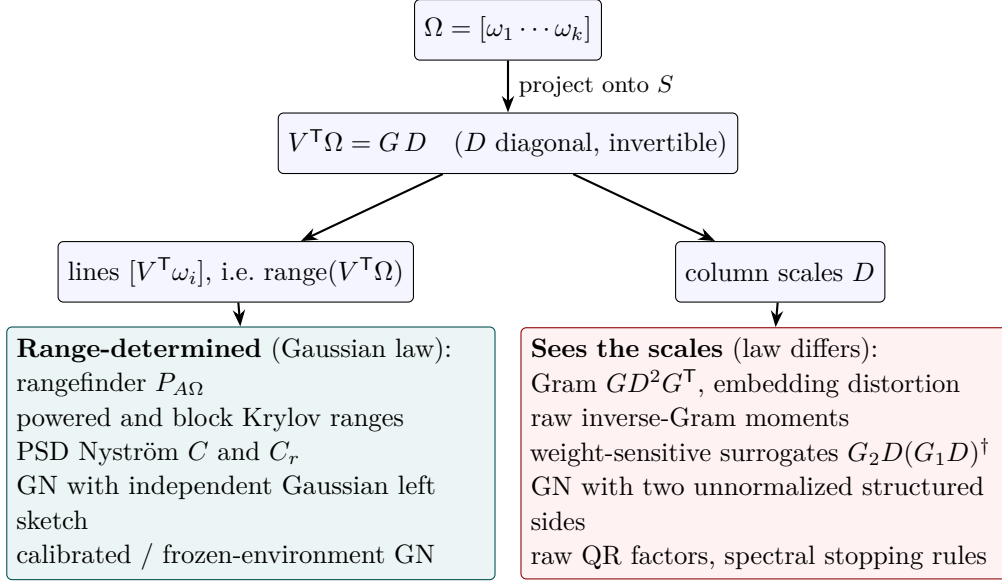


Figure 2: What is and is not range-determined. Under the criterion of Theorem A the lines have their Gaussian law; the scales D do not. Outputs on the left have exactly the dense-Gaussian law; outputs on the right depend on D and generally do not.

Generalized Nyström. Let Q be an orthonormal basis of range Y , $Y = A\Omega = QT$ with T of full row rank. If $\Psi^T Q$ has full column rank then $(\Psi^T QT)^\dagger = T^\dagger (\Psi^T Q)^\dagger$, so

$$\hat{A}_{\text{GN}} = Q(\Psi^T Q)^\dagger \Psi^T A, \quad (3.3)$$

which depends on Ω only through range $Y = AV \text{ range}(V^T \Omega)$. $\square$

Outputs that are *not* range-determined include the sketch Gram matrix $\Omega^T A^T A \Omega$, embedding distortions $\sup_{x \in S} |||\Omega^T x||^2 / \|x\|^2 - 1|$, raw inverse-Gram moments, raw QR factors, stopping rules or regularization chosen from unnormalized sketch singular values, Krylov spaces that include the raw columns of Ω , and generalized Nyström with an unnormalized structured left sketch. Figure 2 summarizes the distinction.

3.2 Proof of the criterion

Proof of Theorem A. (iii) $\Rightarrow$ (i). For a dense Gaussian probe γ , $V^T \gamma \sim N(0, I_R)$ is nonzero almost surely, and its line is uniformly distributed on $\mathbb{RP}^{R-1}$, i.e. has the unique $O(R)$ -invariant probability law, by rotation invariance. Under (iii), the lines $[V^T \omega_1], \dots, [V^T \omega_k]$ are therefore iid with the same law as $[V^T \gamma_1], \dots, [V^T \gamma_k]$. The subspace $\text{range}(V^T \Omega) = \sum_i [V^T \omega_i]$ is a Borel function of this k -tuple of lines. So the two ranges have the same law in $\mathcal{G}_R$, for every k . By Definition 2.1 and independence of the auxiliary randomness, every range-determined output has its Gaussian law.

(i) $\Rightarrow$ (ii). Take $k = 1$ in (3.1): $\text{range}(A\omega) = AV \text{ range}(V^T \omega)$.

(ii) $\Rightarrow$ (iii). Let A have row space S . Then AV is injective and $A\omega = AV(V^T \omega)$. For the Gaussian probe, $\text{range}(A\gamma)$ is a line almost surely. Equality of laws forces $\text{range}(A\omega)$ to be a line almost surely, hence $V^T \omega \neq 0$ almost surely. The map $\varphi: \mathbb{RP}^{R-1} \rightarrow \mathbb{RP}^{N_L-1}$, $\ell \mapsto AV\ell$, is continuous and injective on a compact space, hence a homeomorphism onto its compact image. Since $\varphi([V^T \omega])$ and $\varphi([V^T \gamma])$ have the same law, applying the Borel map φ^{-1} on the image

shows that $[V^\top \omega]$ and $[V^\top \gamma]$ have the same law, the uniform one. The argument used only one A ; since (iii) does not involve A , the equivalence holds for one and for every such A . $\square$

Corollary 3.2 (Coupling and joint statements). *Under the conditions of Theorem A, on a suitable probability space $V^\top \Omega = GD$, where $G \in \mathbb{R}^{R \times k}$ has iid $N(0, 1)$ entries and D is an almost surely invertible diagonal matrix. Consequently, for every finite family of fixed maps B_j annihilating $S^\perp$, jointly, $\text{range}(B_j \Omega) = \text{range}(B_j VG)$ and the range of any horizontal concatenation equals that of the corresponding concatenation with $\Gamma = VG + V_\perp G_\perp$, a dense Gaussian matrix.*

Proof. Put $u = V^\top \omega \neq 0$ and $\theta = u/\|u\|$. Draw an independent uniform sign ϵ and an independent radius $\rho \sim \chi_R$. Since $[\theta]$ is uniform on $\mathbb{RP}^{R-1}$ and ϵ is an independent symmetric sign, $\epsilon\theta$ is uniform on the sphere; with the independent radius, $g = \epsilon\rho\theta \sim N(0, I_R)$. Then $u = (\epsilon\|u\|/\rho)g$. Do this independently for each column. The joint statement follows from $B_j = B_j V V^\top$ and $B_j \Gamma = B_j VG$; an invertible right factor, or a block diagonal one with a copy of D per block, preserves column ranges. $\square$

In this coupling D need not be independent of G . Under the conditional Gaussian mechanism of Section 4 it is.

4 A checkable sufficient mechanism

Projective uniformity is the exact condition, but it is rarely checked directly. The following mechanism is.

Definition 4.1 (Conditional scalar Gaussianity). A probe ω is *conditionally scalar Gaussian* on S if there is a σ -field $\mathcal{F}$ and an $\mathcal{F}$ -measurable random variable w with $w > 0$ almost surely such that

$$V^\top \omega \mid \mathcal{F} \sim N(0, wI_R). \quad (4.1)$$

For a sketch we require that the pairs $(\omega_i, \mathcal{F}_i)$ be independent across columns i .

Theorem 4.2 (Mechanism). *Assume Definition 4.1, including independence of the pairs $(\omega_i, \mathcal{F}_i)$ across i . Then $g_i = V^\top \omega_i / \sqrt{w_i}$ are iid $N(0, I_R)$ and the family (g_i) is independent of (w_i) . Hence*

$$V^\top \Omega = GD, \quad G \text{ iid } N(0, 1), \quad D = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_k}), \quad G \perp D, \quad (4.2)$$

and the conditions of Theorem A hold.

Proof. Conditionally on $\mathcal{F}_i$, $g_i \sim N(0, I_R)$, a law that does not depend on $\mathcal{F}_i$. For Borel E and $F \in \mathcal{F}_i$, $\mathbb{P}\{g_i \in E, F\} = \mathbb{E}[\mathbf{1}_F \mathbb{P}(g_i \in E \mid \mathcal{F}_i)] = \mathbb{P}\{N(0, I_R) \in E\} \mathbb{P}(F)$. So g_i is independent of $\mathcal{F}_i$, in particular of w_i . Independence of the pairs across i gives the joint statement. Finally $V^\top \omega_i = \sqrt{w_i} g_i$ is nonzero almost surely and $[V^\top \omega_i] = [g_i]$ is uniform on $\mathbb{RP}^{R-1}$. $\square$

The column-independence hypothesis is used exactly once, in the joint statement. It matters for designs that share randomness between columns, such as block tensor-train sketches with shared cores; those require a separate joint argument. (Frozen environments, used later, are handled by conditioning on the shared environment, Remark 8.2.)

Remark 4.3 (The mechanism is strictly stronger than the criterion). Equation (4.1) forces $V^\top \omega = \sqrt{w} g$ with g independent of w . Hence $\|V^\top \omega\|$ is a scale mixture of χ_R and is independent of the direction. A probe uniform on the unit sphere of S has constant radius and violates (4.1), and so does $\rho(\theta)\theta$ with a direction-dependent radius. Both satisfy Theorem A.

Corollary 4.4 (Nonzero rescalings, spherical factors). *If ω satisfies the conditions of Theorem A and c is a random scalar that is nonzero almost surely, possibly dependent on ω , then $c\omega$ satisfies them too. In particular, if a Gaussian product probe $g_1 \otimes \cdots \otimes g_q$ satisfies them, so does the spherical product probe $\bigotimes_j \rho_j g_j / \|g_j\|$ for any fixed radii $\rho_j > 0$.*

Proof. $[V^\top(c\omega)] = [V^\top \omega]$, and $V^\top(c\omega) \neq 0$ almost surely. The spherical probe is $c\omega$ with $c = \prod_j \rho_j / \|g_j\|$. $\square$

Remark 4.5 (The complex case). Everything above holds over $\mathbb{C}$ with V^* in place of $V^\top$, unitary invariance in place of orthogonal invariance, the complex projective space $\mathbb{CP}^{R-1}$, and the complex Grassmannian. The mechanism becomes $V^* \omega \mid \mathcal{F} \sim \mathcal{CN}(0, wI_R)$. Complex common-Schmidt product supports and the single-core common-Schmidt tensor-train family are covered after replacing orthonormality by Hermitian orthonormality. For other supports the Hermitian conditional covariance must be checked directly. The real Hurwitz test $\text{sym}(M_a^\top M_c) = 0$ does not imply $M_a^* M_c = 0$, so its real examples do not automatically transfer to complex factors. Spherical factors follow from the complex version of Corollary 4.4. The only change in the Gaussian constants of Section 7 is the pseudoinverse moment: for an $r \times k$ standard complex Gaussian G_1 ($\mathbb{E}|g_{ij}|^2 = 1$) and $k > r$, $\mathbb{E}\|G_1^\dagger\|_F^2 = r/(k-r)$, the complex inverse-Wishart mean.

5 Certified families

We use one elementary fact repeatedly: *a nonzero real polynomial on $\mathbb{R}^d$ has a Lebesgue-null zero set.* (Induction on d : outside the null set of lower variables where all coefficients of the top variable vanish, a nonzero univariate polynomial has finitely many roots; Fubini finishes.) Consequently a polynomial in Gaussian variables that is nonzero with positive probability is nonzero almost surely, and a polynomial identity that holds almost surely holds identically.

5.1 Gaussian product probes

Let $\omega = g \otimes h$ with $g \sim N(0, I_{n_1})$ and $h = h_2 \otimes \cdots \otimes h_q$, where all factors are independent standard Gaussian vectors, and put $n' = \prod_{j \geq 2} n_j$. Reshape $v \in \mathbb{R}^{n_1} \otimes \mathbb{R}^{n'}$ as $M_v \in \mathbb{R}^{n_1 \times n'}$ so that $v^\top(g \otimes h) = g^\top M_v h$. Let $L_h = I_{n_1} \otimes h \in \mathbb{R}^{N \times n_1}$, so that $L_h g = g \otimes h$ and $L_h^\top v = M_v h$. Conditionally on h , $V^\top \omega = V^\top L_h g$ is centered Gaussian with covariance $V^\top L_h L_h^\top V$, whose (a, c) entry is $h^\top M_a^\top M_c h$, where $M_a = M_{u_a}$ for the columns u_a of V .

Definition 5.1 (Conditionally Gaussian product class). The support S belongs to the *conditionally Gaussian product class* if $V^\top L_h L_h^\top V = w(h)I_R$ with $w(h) > 0$ for almost every h ; that is, (4.1) holds with $\mathcal{F} = \sigma(h_2, \dots, h_q)$. The same definition applies with any other factor in the role of g .

Proposition 5.2 (Common-Schmidt family, any order). *Suppose the orthonormal basis of S has the form*

$$u_a = \sum_{b=1}^{\ell} \sqrt{\lambda_b} e_{a,b} \otimes f_b, \quad a = 1, \dots, R, \quad (5.1)$$

where $\lambda_b > 0$, $\sum_b \lambda_b = 1$, the entire family $\{e_{a,b}\}_{a,b} \subset \mathbb{R}^{n_1}$ is orthonormal, and $f_1, \dots, f_\ell$ are orthonormal tensors in $\mathbb{R}^{n'}$ (not necessarily of rank one). Then S belongs to the conditionally Gaussian product class with

$$w(h) = \sum_b \lambda_b \langle f_b, h \rangle^2. \quad (5.2)$$

Necessarily $R\ell \leq n_1$ and $\ell \leq n'$.

Proof. $\langle u_a, g \otimes h \rangle = \sum_b \sqrt{\lambda_b} \langle e_{a,b}, g \rangle \langle f_b, h \rangle$. The coefficients $\langle e_{a,b}, g \rangle$ are iid $N(0, 1)$, and different a use disjoint sets of them, so the conditional covariance is $\delta_{ac} \sum_b \lambda_b \langle f_b, h \rangle^2$. Rank-one tensors span $\mathbb{R}^{n'}$, so $h \mapsto \langle f_b, h \rangle$ is a nonzero polynomial in the factor coordinates; hence $w(h) > 0$ almost surely. The dimension bounds follow from orthonormality. $\square$

Choosing coordinate vectors for $e_{a,b}$ and f_b and any positive eigenvalues μ_a gives inputs $A = \sum_a \mu_a u_a u_a^\top$ in the class, with no restriction on spectral decay. For $\ell = 1$ this is $A = A_1 \otimes f f^\top$ with a fixed unit tensor f and PSD A_1 on the first factor; for $\ell > 1$ the eigenvectors are entangled. The Schmidt weights λ_b describe eigenvectors and must not be confused with the eigenvalues μ_a .

5.2 An algebraic test for two factors, and a Hurwitz family

Proposition 5.3 (Algebraic test). *Let $q = 2$ and let $u_1, \dots, u_R$ be orthonormal. Then S belongs to the conditionally Gaussian product class if and only if there is $S_0 \succeq 0$ with $\text{tr } S_0 = 1$ such that*

$$\text{sym}(M_a^\top M_c) = \delta_{ac} S_0 \quad (1 \leq a, c \leq R), \quad \text{sym } M = \frac{1}{2}(M + M^\top). \quad (5.3)$$

Then $w(h) = h^\top S_0 h$. Equivalently, in basis-free form, $M_v^\top M_v = \|v\|^2 S_0$ for every $v \in S$.

Proof. The conditional covariance entry $h^\top M_a^\top M_c h$ depends only on the symmetric part of $M_a^\top M_c$, which proves sufficiency. Conversely, if $h^\top (M_a^\top M_c - \delta_{ac} M_1^\top M_1) h = 0$ almost surely, the polynomial vanishes identically, so its symmetric coefficient matrix vanishes; put $S_0 = \text{sym}(M_1^\top M_1) = M_1^\top M_1 \succeq 0$, with $\text{tr } S_0 = \|M_1\|_F^2 = 1$. Positivity of $h^\top S_0 h$ almost surely follows because $S_0 \neq 0$. For the basis-free form expand $M_v = \sum_a c_a M_a$; the cross terms are the symmetric parts in (5.3). Conversely, apply the basis-free identity to u_a, u_c and $u_a + u_c$ and polarize. $\square$

Example 5.4 (A family outside the common-Schmidt form). Let $n_1 = n_2 = 2$ and

$$M_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad M_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then $M_1^\top M_1 = M_2^\top M_2 = I_2/2$ and $\text{sym}(M_1^\top M_2) = 0$, so (5.3) holds with $S_0 = I_2/2$. Both M_a have Schmidt rank two, and a common-Schmidt representation with $R = \ell = 2$ would need four orthonormal vectors in $\mathbb{R}^2$. More generally, when $S_0 = I_{n_2}/n_2$ the matrices $L_a = \sqrt{n_2} M_a$ satisfy $L_a^\top L_c + L_c^\top L_a = 2\delta_{ac} I$, equivalently $\|\sum_a x_a L_a y\|^2 = \|x\|^2 \|y\|^2$: these are the classical Hurwitz–Radon sum-of-squares identities [9]. Square families exist exactly up to the Radon–Hurwitz number, e.g. $R = 4$ for $n_1 = n_2 = 4$ (quaternions) and $R = 8$ for $n_1 = n_2 = 8$ (octonions). The identities are classical; here they provide a certificate of exact Gaussian range laws.

Remark 5.5 (Exchanging the factors). Conditioning on g instead of h gives the mirror test $\text{sym}(M_a M_c^\top) = \delta_{ac} S'_0$ with $w = g^\top S'_0 g$; for example $u_a = f \otimes e_a$. For two factors the certified class is the union of the two tests; at higher order any single factor may be left unconditioned.

5.3 A single-core tensor-train family

Let there be q sites with physical dimensions d_j , bond dimensions $\chi_0 = \chi_q = 1$ and $\chi_j = \chi$ for $0 < j < q$, and independent cores $H_j \in \mathbb{R}^{\chi_{j-1} \times d_j \times \chi_j}$ with iid $N(0, 1)$ entries. Define

$$\omega(i_1, \dots, i_q) = \chi^{-(q-1)/2} \sum_{\alpha_1, \dots, \alpha_{q-1}} \prod_{j=1}^q H_j(\alpha_{j-1}, i_j, \alpha_j), \quad \alpha_0 = \alpha_q = 1, \quad (5.4)$$

with all cores regenerated independently for each column [15, 3]. Averaging a product of two entries forces all physical and bond indices to match; the χ^{q-1} surviving bond paths cancel the normalization, so $\mathbb{E}\omega\omega^\top = I$. Fix a site t and condition on the other cores. After a fixed permutation of tensor factors,

$$\omega = \sum_{\alpha=1}^{\chi_{t-1}} \sum_{\beta=1}^{\chi_t} C_{\alpha\beta} \otimes g_{\alpha\beta}, \quad g_{\alpha\beta} = H_t(\alpha, :, \beta), \quad (5.5)$$

where the tensors $C_{\alpha\beta}$ on the other sites absorb the normalization and the $g_{\alpha\beta}$ are iid $N(0, I_{d_t})$.

Proposition 5.6 (Single-core common-Schmidt family). *Suppose S has an orthonormal basis $u_a = \sum_{b=1}^\ell \sqrt{\lambda_b} X_b \otimes \phi_{a,b}$, with $\lambda_b > 0$, $\sum_b \lambda_b = 1$, orthonormal tensors X_b on the other sites, and the entire family $\{\phi_{a,b}\}$ orthonormal in $\mathbb{R}^{d_t}$. Then (4.1) holds conditionally on the other cores, with $w = \sum_{\alpha,\beta,b} \lambda_b \langle X_b, C_{\alpha\beta} \rangle^2 > 0$ almost surely. Necessarily $R\ell \leq d_t$.*

Proof. By (5.5), $\langle u_a, \omega \rangle = \sum_{\alpha,\beta,b} \sqrt{\lambda_b} \langle X_b, C_{\alpha\beta} \rangle \langle \phi_{a,b}, g_{\alpha\beta} \rangle$, which is conditionally Gaussian with covariance $\sum_{\alpha\beta} \sum_{b,c} \sqrt{\lambda_b \lambda_c} \langle X_b, C_{\alpha\beta} \rangle \langle X_c, C_{\alpha\beta} \rangle \langle \phi_{a,b}, \phi_{a',c} \rangle = \delta_{aa'} w$. The function w is a nonnegative polynomial in the other cores, and $\mathbb{E}w = \mathbb{E}\langle u_a, \omega \rangle^2 = 1$ by isotropy, so w is not identically zero and is positive almost surely. $\square$

Orthonormal product tensors X_b and distinct coordinate vectors $\phi_{a,b}$ give concrete PSD inputs with Gaussian Nyström laws, for every χ and q . At fixed d_t this construction has $R \leq d_t/\ell$; conditioning on one core always bounds the conditional covariance rank by $d_t \chi_{t-1} \chi_t$.

6 The rank ceiling

The families of Section 5 all have small rank. This is not an artifact of the constructions. We use the following standard fact [8, Corollary 6.11]: *if $f: M \rightarrow N$ is a C^1 map between smooth manifolds with $\dim M < \dim N$, then $f(M)$ has measure zero in N* (for any smooth positive density on N).

Proof of Theorem B. Let $\Sigma = \{x_1 \otimes \dots \otimes x_q\}$ be the cone of rank-one tensors and $M = S^{n_1-1} \times \dots \times S^{n_q-1}$, a manifold of dimension δ . Every nonzero $x \in \Sigma$ equals $c \xi_1 \otimes \dots \otimes \xi_q$ with $c \neq 0$ and $(\xi_j) \in M$. Let $U \subseteq M$ be the open set where $V^\top(\xi_1 \otimes \dots \otimes \xi_q) \neq 0$, and $\varphi: U \rightarrow \mathbb{R}^{R-1}$, $\xi \mapsto [V^\top(\xi_1 \otimes \dots \otimes \xi_q)]$, a smooth map.

(i) Under Theorem A, $V^\top \omega \neq 0$ almost surely and $[V^\top \omega] = \varphi(\xi) \in \varphi(U)$, where ξ is the normalized factor tuple. If $\delta < R - 1$, then $\varphi(U)$ is null in $\mathbb{R}^{R-1}$, contradicting uniformity of $[V^\top \omega]$. Hence $R \leq \delta + 1$. Under conditional scalar Gaussianity the conditions of Theorem A hold (Theorem 4.2), so the same bound applies. Attainment is Example 6.1 below.

(ii) Fix $k < R$. For the Gaussian sketch, $\text{range}(V^\top \Gamma)$ lies in $\text{Gr}(k, R)$ almost surely and has the $O(R)$ -invariant law, which has a smooth positive density. If the structured range has the same

law, it is k -dimensional almost surely; then all columns are nonzero, their normalized factor tuples lie in U , and $\text{range}(V^\top \Omega) = \Phi(\xi^{(1)}, \dots, \xi^{(k)})$, where $\Phi(\xi^{(1)}, \dots, \xi^{(k)}) = \varphi(\xi^{(1)}) + \dots + \varphi(\xi^{(k)})$ is smooth on the open subset of U^k where these lines are independent. That subset has dimension $k\delta$ and $\dim \text{Gr}(k, R) = k(R - k)$. If $R > k + \delta$ then $k\delta < k(R - k)$, the image of Φ is null, and we reach a contradiction. Hence $R \leq k + \delta$. Nothing about the joint law of the columns was used.

(iii) In the conditionally Gaussian product class, $V^\top L_h L_h^\top V = w(h)I_R$ with $w(h) > 0$, so the $R \times n_1$ matrix $V^\top L_h$ has rank R , and $R \leq n_1$. $\square$

Example 6.1 (The ceiling is attained). Let $z_0 \in \mathbb{R}$ and $z_j \in \mathbb{R}^{n_j-1}$, $j = 1, \dots, q$, be independent standard Gaussian, and put

$$x_1 = (z_1, z_0), \quad x_j = (z_j/z_0, 1) \quad (j \geq 2), \quad \omega = x_1 \otimes \dots \otimes x_q.$$

Let H be the “hook” of multi-indices $(i_1, \dots, i_q)$ with at most one coordinate $i_j < n_j$ and all others equal to their maximum. Then $|H| = \delta + 1$, $\omega(n_1, \dots, n_q) = z_0$, the entries with $i_1 < n_1$ are the coordinates of z_1 , and the entries with $i_j < n_j$ for $j \geq 2$ equal $z_0 \cdot z_j(i_j)/z_0 = z_j(i_j)$. With V the coordinate isometry onto $\mathbb{R}^H$, $V^\top \omega = (z_0, z_1, \dots, z_q) \sim N(0, I_{\delta+1})$: the mechanism holds with $w \equiv 1$ on a support of dimension $\delta + 1$. The factors share z_0 and are therefore dependent; the ratios in the later factors are heavy-tailed. This attains the ceiling under the arbitrary joint factor laws allowed in Theorem B(i), and does not establish attainment for independent factors. For independent Gaussian factors and the conditional mechanism, part (iii) gives the smaller ceiling n_1 .

Remark 6.2 (What the ceiling means). With base dimension two ($n_j = 2$, $n = 2^q$) we get $\delta = q$ and $R \leq \log_2 n + 1$; for $n = 2^{20}$, at most 21. For $k \geq R$ the Gaussian rangefinder is exact and the question is vacuous. So the class-specific theorems of Sections 7–10 have content only in the window $r + 1 < k < R \leq \delta + 1$, and for the conditionally Gaussian product class in $r + 1 < k < R \leq n_1$. They are exact identifications, and they should be framed as such: the class is explicit and checkable, but it is not a large class of inputs.

Remark 6.3 (Tensor trains: growing rank is not excluded). The probe (5.4) lies in the tensor-train variety $\mathcal{T}_\chi$ of tensors with TT ranks at most χ . It is the image of the core space, of dimension $D_\chi = \sum_j \chi_{j-1} d_j \chi_j$, under a polynomial map. The proof of part (i), applied to this map (whose line is unchanged when the first core is multiplied by a nonzero scalar, so the effective parameter count is $D_\chi - 1$), gives $R \leq D_\chi$ under the conditions of Theorem A; with the semialgebraic dimension of $\mathcal{T}_\chi$ in place of the parameter count, $R \leq \dim \mathcal{T}_\chi \leq D_\chi$. This bound grows with χ and q , so the dimension count does *not* rule out Gaussian tensor-train probes with exact Gaussian range laws on supports of growing rank. Proposition 5.6 is the only tensor-train family proved here, and it has $R \leq d_t/\ell$ at fixed physical dimension. We make no claim of a growing-rank tensor-train theorem; see Section 13.

7 Gaussian approximation bounds on the class

Equality of laws transfers every correctly scoped Gaussian theorem about a range-determined output. We give complete proofs of the two most used bounds, classical in the Gaussian setting [6, 17], to make visible that random column scales and tensor dimensions contribute no extra factor.

Lemma 7.1 (Gaussian pseudoinverse moment). *Let $G_1 \in \mathbb{R}^{r \times k}$ have iid $N(0, 1)$ entries and $k > r + 1$. Then $\mathbb{E} \|G_1^\dagger\|_F^2 = r/(k - r - 1)$.*

Proof. The rows are independent almost surely and $\|G_1^\dagger\|_F^2 = \text{tr}(G_1 G_1^\top)^{-1}$. By the Schur complement, the i -th diagonal entry of $(G_1 G_1^\top)^{-1}$ is the reciprocal of the squared distance from row i to the span of the other $r - 1$ rows. Conditionally on those rows this squared distance is χ_ν^2 with $\nu = k - r + 1$, and $\mathbb{E}(\chi_\nu^2)^{-1} = 1/(\nu - 2)$ by the gamma integral. Sum over i . $\square$

Theorem 7.2 (Range-finder). *Let the row space S of A satisfy Theorem A, and let $r + 1 < k \leq R$. Then $P_{A\Omega}$ has its Gaussian law and*

$$\mathbb{E}\|A - P_{A\Omega}A\|_F^2 \leq \left(1 + \frac{r}{k - r - 1}\right) \tau_r^{(2)}(A). \quad (7.1)$$

If $k \geq R$ the range-finder is exact almost surely.

Proof. By Lemma 3.1(a) and Theorem A we may replace $V^\top \Omega$ by a Gaussian G .

Step 1: construct a candidate inside the sketch range. With the compact SVD $A = U\Sigma V^\top$, split $G = [G_1; G_2]$ after r rows and $\Sigma = \text{diag}(\Sigma_1, \Sigma_2)$. The range of $Y = U\Sigma G$ contains the columns of

$$Y G_1^\dagger = U \begin{bmatrix} \Sigma_1 \\ \Sigma_2 G_2 G_1^\dagger \end{bmatrix}.$$

Using this matrix times the leading right singular vectors as a candidate, and since orthogonal projection is optimal among matrices with columns in $\text{range } Y$, we obtain

$$\|A - P_Y A\|_F^2 \leq \|\Sigma_2\|_F^2 + \|\Sigma_2 G_2 G_1^\dagger\|_F^2.$$

Step 2: average the cross term. Conditionally on G_1 , G_2 is an independent Gaussian matrix and

$$\mathbb{E}[\|\Sigma_2 G_2 G_1^\dagger\|_F^2 \mid G_1] = \|\Sigma_2\|_F^2 \|G_1^\dagger\|_F^2.$$

Apply Lemma 7.1. For $k \geq R$, G has full row rank almost surely. $\square$

Theorem 7.3 (PSD Nyström, including fixed rank). *Let $A \succeq 0$ have range S satisfying Theorem A. Then $C = A\Omega(\Omega^\top A\Omega)^\dagger \Omega^\top A$ and its best rank- r truncation C_r have their Gaussian laws, and for $r + 1 < k \leq R$,*

$$\mathbb{E} \text{tr}(A - C) \leq \mathbb{E} \text{tr}(A - C_r) \leq \left(1 + \frac{r}{k - r - 1}\right) \tau_r(A). \quad (7.2)$$

If $k \geq R$ then $C = A$ almost surely and C_r is a best rank- r approximation.

Proof. By (3.2), $0 \preceq C_r \preceq C \preceq A$, and by Theorem A we may take $V^\top \Omega = G$ Gaussian.

Step 1: identify an r -dimensional graph inside the sampled range. In the positive eigenbasis $A = \text{diag}(\Lambda_1, \Lambda_2)$, the range of $A^{1/2}G$ contains the graph subspace spanned by $\begin{bmatrix} I \\ F \end{bmatrix}$ with

$$F = \Lambda_2^{1/2} G_2 G_1^\dagger \Lambda_1^{-1/2}.$$

Let P_F be its projector; its leading block is $(I + F^\top F)^{-1}$. The sum of the r largest eigenvalues of $C = A^{1/2} P A^{1/2}$ equals that of $P A P$ and is at least $\text{tr}(P_F A)$ by Ky Fan's principle, since $\text{range } P_F \subseteq \text{range } P$ has dimension r .

Step 2: bound the lost trace. The trailing contribution to $\text{tr}(P_F A)$ is nonnegative and $I - (I + F^\top F)^{-1} \preceq F^\top F$, so

$$\begin{aligned} \text{tr}(A - C_r) &\leq \tau_r(A) + \text{tr}[\Lambda_1(I - (I + F^\top F)^{-1})] \\ &\leq \tau_r(A) + \|\Lambda_2^{1/2} G_2 G_1^\dagger\|_F^2. \end{aligned}$$

Conditional Gaussian averaging over G_2 and Lemma 7.1 finish the proof; exactness for $k \geq R$ is as in Theorem 7.2. $\square$

Thus $k = r + 1 + \lceil r/\varepsilon \rceil$ columns suffice for expected $(1 + \varepsilon)\tau_r(A)$ error. The Gaussian fixed-rank spectral bound $\mathbb{E}\|A - C_r\| \leq \lambda_{r+1}(A) + \frac{r}{k-r-1}\tau_r(A)$ of [17, Theorem 4.1], the Gaussian power-method bounds of [6], the block Krylov bounds of Musco and Musco [10], and all Gaussian tail bounds for these outputs transfer verbatim by Theorem A and Lemma 3.1.

7.1 Consequence for Khatri–Rao Nyström

The range criterion avoids weighted inverse-Gram estimates. To make the comparison self-contained, we prove a conservative weighted bound for the two-factor Gaussian model. This bound also supplies the raw left-sketch estimate in Proposition 9.6. It is not an embedding theorem.

Theorem 7.4 (A weighted inverse bound). *Let $r \geq 1$, let $\Xi \in \mathbb{R}^{k \times r}$ be standard Gaussian, and let $w_i = \sum_b \lambda_b z_{ib}^2$, where $\lambda_b > 0$, $\sum_b \lambda_b = 1$, and the z_{ib} are iid standard Gaussian, independent of Ξ . Set $W = \text{diag}(w_i)$ and $S_k = k^{-1}\Xi^\top W \Xi$. For $k \geq 4(r + 1)$,*

$$\mathbb{E}S_k^{-1} \preceq C_{\text{new}}I_r, \quad C_{\text{new}} = \frac{112}{3\pi} - \frac{4}{3} < 10.551.$$

If $\Omega_1 = \Xi^\top W^{1/2}$ and $\Omega_2 = \Xi_2^\top W^{1/2}$ with an independent standard Gaussian Ξ_2 , then

$$\mathbb{E}\|\Lambda_2^{1/2}\Omega_2\Omega_1^\dagger\|_{\text{F}}^2 \leq 3C_{\text{new}}\frac{r}{k}\text{tr}\Lambda_2 \leq 31.651\frac{r}{k}\text{tr}\Lambda_2.$$

Proof. Step 1: reduce to one squared Gaussian weight. Put $H_b = \sum_i z_{ib}^2 \xi_i \xi_i^\top$, where $\xi_i^\top$ is row i of Ξ . Then $kS_k = \sum_b \lambda_b H_b$. All these matrices are positive definite almost surely. Matrix inversion is convex in the Loewner order, so

$$(kS_k)^{-1} \preceq \sum_b \lambda_b H_b^{-1}.$$

One elementary proof of this convexity is to maximize $2x^\top y - y^\top H y$ over y for each fixed x : its value is $x^\top H^{-1}x$, a supremum of affine functions of H . The H_b have the same marginal law. It therefore suffices to take $w_i = z_i^2$.

Step 2: keep the largest half of the weights. Let $s = \lceil k/2 \rceil$, and let v be the smallest weight among the largest s . If J is their index set, then

$$H = \Xi^\top W \Xi \succeq v \sum_{i \in J} \xi_i \xi_i^\top, \quad \mathbb{E}H^{-1} \preceq \frac{\mathbb{E}v^{-1}}{s - r - 1}I_r.$$

The second statement follows by conditioning on the weights: the selected Gaussian rows remain independent standard Gaussian. Their Gram matrix has the inverse mean $I_r/(s - r - 1)$ by Lemma 7.1 and rotation invariance. The denominator is positive under the stated sample budget.

Step 3: control the order statistic. Write F_1 for the χ_1^2 distribution function. Comparing derivatives in $x = \sqrt{v}$, using $\log(1 + x^2/3) \leq x^2/3$, gives

$$F_1(v) \leq \sqrt{\frac{2}{\pi}} \frac{\sqrt{v}}{\sqrt{1 + v/3}}, \quad v^{-1} \leq \frac{2}{\pi F_1(v)^2} - \frac{1}{3}.$$

The variable $U = F_1(v)$ is the $(k - s + 1)$ -st order statistic of k independent uniforms. Counting the uniforms below u gives its Beta density $u^{k-s}(1-u)^{s-1}/B(k-s+1, s)$. The beta integral yields

$$\mathbb{E}U^{-2} = \frac{k(k-1)}{(k-s)(k-s-1)}, \quad k\mathbb{E}H^{-1} \preceq \frac{k}{s-r-1} \left[\frac{2k(k-1)}{\pi(k-s)(k-s-1)} - \frac{1}{3} \right] I_r.$$

For $k = 2m$, $m \geq 4$, the first factor is at most 4, and $\mathbb{E}U^{-2} = 4 + 2/(m-1) \leq 14/3$. This gives C_{new} . For $k = 2m+1$, $m \geq 4$ and $m \geq 2(r+1)$, the coefficient is at most

$$\frac{4m+2}{m+2} \left[\frac{2}{\pi} \left(4 + \frac{6}{m-1} \right) - \frac{1}{3} \right].$$

This expression decreases for $m \geq 4$: after subtracting its limit $32/\pi - 4/3$, the remainder is $[2\pi m + 72 - 2\pi]/[\pi(m-1)(m+2)]$, which is decreasing. Its value at $m = 4$ is $36/\pi - 1 < C_{\text{new}}$.

Step 4: bound the surrogate. Conditional averaging over Ξ_2 reduces the desired bound to

$$\text{tr } \Lambda_2 \mathbb{E} \text{tr}(H^{-1} \Xi^\top W^2 \Xi H^{-1}).$$

Lemma 9.5, applied row by row after adding ηI to H , bounds this by $(\mathbb{E}w_i^2) \text{tr } \Lambda_2 \mathbb{E} \text{tr } H^{-1}$; the details of the same calculation are given in Proposition 9.6. Since $\mathbb{E}w_i^2 = 1 + 2 \sum_b \lambda_b^2 \leq 3$, the first assertion completes the proof. No part of the weighted inverse estimate uses that proposition. $\square$

Corollary 7.5 (Sharp Khatri–Rao Nyström constant). *If every positive-eigenvalue eigenvector of $A \succeq 0$ has the common-Schmidt form (5.1) with one list of weights and a jointly orthonormal family $\{e_{a,b}\}$, then Gaussian Khatri–Rao Nyström with k columns has exactly the dense-Gaussian output law, and $\mathbb{E} \text{tr}(A - C_r) \leq (1 + \frac{r}{k-r-1}) \tau_r(A)$ for every $k > r+1$. For $k \geq 4(r+1)$ one has $r/(k-r-1) \leq \frac{4}{3} r/k$, so the coefficient improves on the conservative weighted bound in Theorem 7.4 by a factor of more than 23.*

Proof. Proposition 5.2, Theorem 4.2 and Theorem 7.3. For $k \geq 4(r+1)$, $k-r-1 \geq \frac{3}{4}k$. $\square$

The looseness of the weighted route is explained in Proposition 11.2: the surrogate $G_2 D(G_1 D)^\dagger$ is conditionally never smaller than the range representative $G_2 G_1^\dagger$. The weighted inverse bounds remain the right tool for observables that are not range-determined, such as inverse-Gram moments and embeddings.

8 The class is an $n_1 \times n_1$ matrix in disguise

This section works in the *product-query model*: the algorithm may request Az (or $z^\top A$) only for legal rank-one product vectors $z = x_1 \otimes \cdots \otimes x_q$, and receives full response vectors. We show that for the conditionally Gaussian product class this model is as powerful as an ordinary matrix–vector oracle for an $n_1 \times n_1$ matrix.

8.1 Conditional-linear generators and frozen environments

More generally, let the probe be $\omega = L_\eta g$ with $g \sim N(0, I_a)$ independent of an environment η , where the matrix $L_\eta \in \mathbb{R}^{N \times a}$ is known from the sampled environment and each $L_\eta x$ is a legal query. The mechanism (4.1) with $\mathcal{F} = \sigma(\eta)$ reads

$$V^\top L_\eta L_\eta^\top V = w(\eta) I_R, \quad w(\eta) > 0 \text{ a.s.} \quad (8.1)$$

For the conditionally Gaussian product class, $\eta = (h_2, \dots, h_q)$, $h = h_2 \otimes \dots \otimes h_q$, $a = n_1$, and $L_\eta x = x \otimes h$.

Lemma 8.1 (Frozen environments). *In the conditionally Gaussian product class, (8.1) holds for every $(h_2, \dots, h_q)$, with $w(h) = \text{tr}(V^\top L_h L_h^\top V)/R = \|M_{u_1} h\|^2$, and $w(h) > 0$ off a Lebesgue-null algebraic set. We call such h good environments.*

Proof. The entries of $V^\top L_h L_h^\top V - (\text{tr}(V^\top L_h L_h^\top V)/R)I_R$ are polynomials in the coordinates of $h_2, \dots, h_q$ that vanish almost surely, hence identically. The polynomial w is positive almost surely, hence nonzero, and its zero set is null. $\square$

In particular, *any* fixed product environment with $w > 0$ works, at any order q : a random Gaussian product environment is good almost surely, and no entangled eigenvector of the environmental space is needed.

Remark 8.2 (Frozen-environment sketches have Gaussian range laws). If the columns $\omega_i = L_\eta g_i$ share one environment η , independent of iid $g_i \sim N(0, I_a)$, then conditionally on η we have $V^\top \Omega = \sqrt{w(\eta)} G$ with G standard Gaussian, so $\text{range}(V^\top \Omega) = \text{range}(G)$. This conditional law does not depend on η ; hence every range-determined output again has its dense-Gaussian law, for every k , although the columns are not independent.

8.2 One-response calibration

Theorem 8.3 (One supported response determines the scale). *Assume (8.1). Then, for every good η and every nonzero $v \in S$,*

$$\hat{w}_v(\eta) := \frac{\|L_\eta^\top v\|^2}{\|v\|^2} = w(\eta). \quad (8.2)$$

Consequently, the normalized probes $\tilde{\omega}_i = \omega_i / \sqrt{\hat{w}_v(\eta_i)}$ satisfy $V^\top \tilde{\omega}_i \stackrel{\text{iid}}{\sim} N(0, I_R)$, independent of the environments. This remains true when v is computed from an independent sketch, because (8.2) is an identity, not an estimate.

Proof. Write $v = Vc$. Then $\|L_\eta^\top v\|^2 = c^\top V^\top L_\eta L_\eta^\top V c = w(\eta) \|c\|^2$ and $\|c\| = \|v\|$. The identity holds simultaneously for all $v \in S$ on the probability-one set of good environments, and Theorem 4.2 gives the rest. $\square$

A supported vector is already available on the output side: if $A \neq 0$ and ω_0 is a Gaussian product probe, then $A\omega_0 \in \text{range } A$; for PSD A this is the relevant support S . It is nonzero almost surely, since some coordinate is a nonzero polynomial in the factor coordinates (product vectors span the domain). A rectangular row support instead needs a left response $A^\top \psi_0$. The expression $A^\top (A\omega_0)$ is supported but is not free in a product-only oracle, because $A\omega_0$ need not be a legal left query. One response does not *certify* the structural hypothesis; the theorem applies when it holds.

For two factors, $L_h^\top v = M_v h$, and Proposition 5.3 gives the basis-free recovery

$$S_0 = \frac{M_v^\top M_v}{\|v\|^2}, \quad w(h) = h^\top S_0 h, \quad (8.3)$$

from any nonzero $v \in S$. For the common-Schmidt family $S_0 = \sum_b \lambda_b f_b f_b^\top$. Neither the eigenbasis of A nor the Schmidt representation is needed. A well-conditioned two-factor environment is $h_\star = f_\star / \sqrt{\lambda_\star}$ with $(\lambda_\star, f_\star)$ a top eigenpair of S_0 ; then $w(h_\star) = 1$ and $\|h_\star\|^2 = 1/\lambda_\star \leq \text{rank } S_0$.

8.3 One query per matrix–vector product, and the compressed matrix

Proof of Theorem C. Fix a good environment h (Lemma 8.1); write $L = L_h$ and $w = w(h)$, and let $A = V\Lambda V^\top$ with $\Lambda \succ 0$ of size R .

Part (a): recover the scale. This is (8.2).

Part (b): match the support projection with one legal query. For $v = Vc \in S$, the scalar covariance identity implies

$$P_S(LL^\top v/w) = V(V^\top LL^\top V)c/w = Vc = v.$$

Since $A = AP_S$, any query vector with this projection gives the same response as v . In particular,

$$A(LL^\top v/w) = Av, \quad LL^\top v/w = (L^\top v/w) \otimes h.$$

The latter is a legal rank-one product. For two factors its first factor is $M_v h/w$. The vector need not equal v outside the support; equality of its projection is enough.

Part (c): compress without changing the positive spectrum. Put $J = V^\top L / \sqrt{w} \in \mathbb{R}^{R \times n_1}$. By (8.1),

$$JJ^\top = I_R, \quad B = L^\top AL/w = J^\top \Lambda J.$$

The nonzero eigenvalues of $J^\top \Lambda J$ are those of $\Lambda J J^\top = \Lambda$, with multiplicities. Moreover $Bx = L^\top[A(Lx)]/w$, one product query $Lx = x \otimes h$.

Part (d): recover the input by querying a basis of the first factor. Take $Z = L$, i.e. the n_1 queries $e_i \otimes h$. The matrix $V^\top L$ has rank R , so

$$\text{range}(A^{1/2}Z) = A^{1/2}V \text{range}(V^\top L) = \text{range } A^{1/2}.$$

The projector representation (3.2) now gives

$$AL(L^\top AL)^\dagger L^\top A = A.$$

The scale w is read from any nonzero response column $y = A(e_i \otimes h)$ via (a), and $\text{tr } A = \text{tr } \Lambda = \text{tr } B$. Since $V^\top L$ has rank R , $R \leq n_1$.

Part (e): count the simulated powers. For $A \succeq 0$ every column of $A^j \Omega$, $j \geq 1$, lies in S , so by (b) each further multiplication costs one query per column: k queries for $A\Omega$ and k per power, $k(1+t)$ in total through $A^{t+1}\Omega$. The simulated products are exact, so the joint ranges are exactly those of the powered and Krylov sketches, and they have their Gaussian laws whenever $\text{range}(V^\top \Omega)$ does (Lemma 3.1(b)). The rectangular case is Proposition 8.6. $\square$

Corollary 8.4 (Spectral transfers). *In the setting of Theorem C:*

(i) *For every f with $f(0) = 0$, $\text{tr } f(A) = \text{tr } f(B)$. In particular, the effective dimensions agree:*

$$d_{\text{eff}}(A, \rho) = \text{tr}[A(A + \rho I)^{-1}] = d_{\text{eff}}(B, \rho).$$

(ii) *For $\rho > 0$, the determinants differ only by the extra zero spectrum:*

$$\log \det(A + \rho I_N) = (N - n_1) \log \rho + \log \det(B + \rho I_{n_1}).$$

(iii) if $By = \lambda y$, $\|y\| = 1$, and $\lambda > 0$, then $u = A(Ly)/(\lambda\sqrt{w})$ is a unit eigenvector of A for λ , at the cost of one more product query.

Proof. (i) and (ii) follow from (c): A has $N - R$ zero eigenvalues and B has $n_1 - R$. For (iii), a unit eigenvector y of $J^\top \Lambda J$ with $\lambda > 0$ is $y = J^\top c$ with $\Lambda c = \lambda c$, $\|c\| = 1$. Then $P_S(Ly) = V(V^\top L)(L^\top V)c/\sqrt{w} = \sqrt{w} Vc$, so $A(Ly) = \sqrt{w} AVc = \lambda\sqrt{w} Vc$. $\square$

Thus stochastic Lanczos quadrature, trace-function estimation and Gaussian Nyström preconditioning can be run on B with product-only calls; their classical accuracy analyses apply to B unchanged. For example, the effective-dimension Nyström preconditioner analysis of Frangella, Tropp and Udell [5] becomes available on this class. Supported linear solves also cost one query per iteration: conjugate gradients for $(A + \rho I)x = b$ with $b \in S$ stays in S ; for a legal product right-hand side $b \notin S$, use $(A + \rho I)^{-1}b = \rho^{-1}b - \rho^{-1}(A + \rho I)^{-1}Ab$, whose remaining solve has the supported right-hand side Ab .

Remark 8.5 (The Schmidt-expansion route is superseded). For two factors, M_v annihilates $\ker S_0$ for $v \in S$, so $v = \sum_{b=1}^s (M_v f_b) \otimes f_b$ with f_b spanning range S_0 and $s = \text{rank } S_0$; this simulates Av with s product queries, and gives power costs $k(1 + st)$ and $k[1 + t(s_L + s_R)]$. Part (b) needs one query, and it does not reconstruct v itself, only its projection onto S , which is all A sees. At higher order the expansion would require the f_b to be legal products; a frozen product environment removes this obstruction.

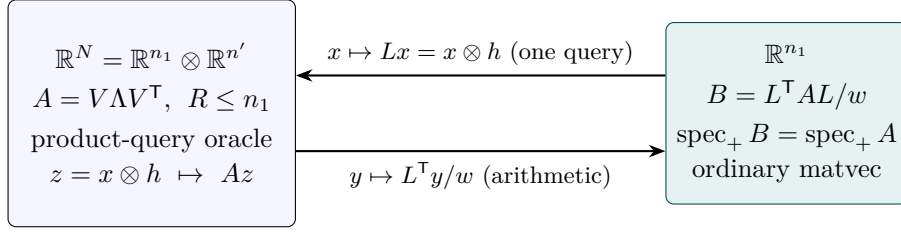
Proposition 8.6 (Rectangular inputs). *Let $A \in \mathbb{R}^{N_L \times N_R}$, and let its column space S_L and row space S_R belong to the conditionally Gaussian product class for left and right product generators $L_h^{(L)}$ and $L_{h'}^{(R)}$ with first-factor dimensions n_1^L and n_1^R . Fix good environments h, h' with scales w_L, w_R . Then:*

- (i) $A^\top y = A^\top (L^{(L)} L^{(L)\top} y / w_L)$ for $y \in S_L$ and $Ax = A(L^{(R)} L^{(R)\top} x / w_R)$ for $x \in S_R$; t cycles $Y \mapsto AA^\top Y$ started from $Y = A\Omega$ cost $k(1 + 2t)$ product queries;
- (ii) $B = L^{(L)\top} A L^{(R)} / \sqrt{w_L w_R} \in \mathbb{R}^{n_1^L \times n_1^R}$ has the same positive singular values as A ;
- (iii) the n_1^R right queries $Y = A L^{(R)}$ and n_1^L left queries $X = L^{(L)\top} A$ recover $A = Y(L^{(L)\top} Y)^\dagger X$ exactly.

Proof. (i) is part (b) of Theorem C applied to $A^\top$ on S_L and to A on S_R ; the columns of Y lie in S_L and those of $A^\top Y$ in S_R , so each cycle costs two queries per column; the scales are read from the first nonzero column of Y and of $A^\top Y$. (ii) With $A = U\Sigma V^\top$, $J_L = U^\top L^{(L)} / \sqrt{w_L}$ and $J_R = V^\top L^{(R)} / \sqrt{w_R}$ have orthonormal rows, and $B = J_L^\top \Sigma J_R$. (iii) range $Y = \text{range } A$ because $V^\top L^{(R)}$ has full row rank, and $L^{(L)\top}$ is injective on S_L because $\|L^{(L)\top} y\|^2 = w_L \|y\|^2$. Formula (3.3) with $\Psi = L^{(L)}$ and Q a basis of range A then returns $QQ^\top A = A$. $\square$

Remark 8.7 (Exact arithmetic and conditioning). All identities in this section are exact-arithmetic statements. A good but unlucky environment may have small $w(h)$, which amplifies rounding error in $L^\top v / w$. For two factors the environment $h_\star$ defined after (8.3) has $w = 1$; at higher order a random Gaussian product environment, or the best of a few, is the practical choice. No such identity is guaranteed for vectors outside S .

good environment h : $V^\top LL^\top V = wI_R$, $w = \|L^\top v\|^2/\|v\|^2$



$$v \in S: \quad Av = A((L^\top v/w) \otimes h) \quad n_1 \text{ queries } e_i \otimes h \text{ recover } A$$

Figure 3: The n_1 -dimensional reduction. One legal product query realizes one product with the compressed matrix B , or one supported product with A .

9 Generalized Nyström

Generalized Nyström [18, 12] approximates a rectangular $A \in \mathbb{R}^{N_L \times N_R}$ from a right sketch $Y = A\Omega$ (k columns) and a left sketch $X = \Psi^\top A$ (p rows) by

$$\hat{A} = Y(\Psi^\top Y)^\dagger X. \quad (9.1)$$

By (3.3), if Q is an orthonormal basis of $\text{range } Y$ and $\Psi^\top Q$ has full column rank, then $\hat{A} = Q(\Psi^\top Q)^\dagger \Psi^\top A$. The left sketch therefore fixes a least-squares *metric*, not only a subspace, and this is where structured left sketches need care. Throughout, $S_L = \text{range } A$ with isometry $U \in \mathbb{R}^{N_L \times R}$, and the right and left sketches are independent.

Lemma 9.1 (Gaussian regression identity). *Let $\text{range } Y \subseteq S_L$ be k -dimensional with orthonormal basis Q , and let $\tilde{\Psi} \in \mathbb{R}^{N_L \times p}$, $p > k + 1$, be independent of Y with $\tilde{\Psi}^\top U$ having iid $N(0, 1)$ entries. Then $\tilde{A} = Q(\tilde{\Psi}^\top Q)^\dagger \tilde{\Psi}^\top A$ satisfies*

$$\mathbb{E}[\|A - \tilde{A}\|_F^2 \mid Q] = \left(1 + \frac{k}{p - k - 1}\right) \|(I - QQ^\top)A\|_F^2. \quad (9.2)$$

Proof. Step 1: split the Gaussian measurements into fitted and residual parts. Complete Q to an orthonormal basis $[Q \ Q_\perp]$ of S_L . Conditionally on Q , $\Gamma_1 = \tilde{\Psi}^\top Q$ and $\Gamma_2 = \tilde{\Psi}^\top Q_\perp$ are independent standard Gaussian matrices, by rotation invariance. Since the columns of A lie in S_L , we have

$$\tilde{\Psi}^\top A = \Gamma_1 Q^\top A + \Gamma_2 F, \quad F = Q_\perp^\top A.$$

Also $\Gamma_1^\dagger \Gamma_1 = I_k$ almost surely.

Step 2: use orthogonality of the two error components. Writing $E = (I - QQ^\top)A = Q_\perp F$, we obtain

$$A - \tilde{A} = E - Q\Gamma_1^\dagger \Gamma_2 F.$$

The two terms are Frobenius-orthogonal, and

$$\mathbb{E}[\|\Gamma_1^\dagger \Gamma_2 F\|_F^2 \mid \Gamma_1] = \|\Gamma_1^\dagger\|_F^2 \|F\|_F^2.$$

Lemma 7.1 applied to $\Gamma_1^\top$ gives $\mathbb{E}\|\Gamma_1^\dagger\|_F^2 = k/(p - k - 1)$, and $\|F\|_F = \|E\|_F$. $\square$

Theorem 9.2 (Structured right sketch, Gaussian left sketch). *Let the row space of A satisfy Theorem A, and let Ψ be an independent dense Gaussian matrix with $p > k + 1$ columns. Then (9.1) has exactly the output law of generalized Nyström with Gaussian sketches on both sides, and for $r + 1 < k$,*

$$\mathbb{E}\|A - \hat{A}\|_{\text{F}}^2 \leq \left(1 + \frac{k}{p - k - 1}\right) \left(1 + \frac{r}{k - r - 1}\right) \tau_r^{(2)}(A). \quad (9.3)$$

Proof. The output is range-determined (Lemma 3.1(d)), so its law is the Gaussian one. Lemma 9.1 with $\tilde{\Psi} = \Psi$ and Theorem 7.2 give (9.3); if $k \geq R$ the error is zero. $\square$

The dense left sketch costs p general left products, which may be unavailable in a product-query model. The next two results make both sides structured.

9.1 Calibrated and frozen-environment reconstruction

Let S_L be in the conditionally Gaussian product class for a left generator, and draw left probes $\psi_i = L_{\eta_i} g_i$, $i = 1, \dots, p$.

Calibrated estimator. With independent environments η_i , choose a nonzero column v of Y (it lies in S_L), compute $w_i = \|L_{\eta_i}^\top v\|^2 / \|v\|^2$ by (8.2), set $D_L = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_p})$, and return

$$\hat{A}_{\text{cal}} = Y(D_L^{-1} \Psi^\top Y)^\dagger D_L^{-1} \Psi^\top A. \quad (9.4)$$

The rescaling acts on stored rows and needs no further queries.

Frozen-environment estimator. Draw one good environment η_L , independent of everything else, and set $\psi_i = L_{\eta_L} g_i$. Return the unmodified formula $\hat{A}_{\text{fr}} = Y(\Psi^\top Y)^\dagger \Psi^\top A$.

Theorem 9.3 (Structured generalized Nyström). *In both constructions, conditionally on Q ,*

$$\mathbb{E}[\|A - \hat{A}\|_{\text{F}}^2 \mid Q] = \left(1 + \frac{k}{p - k - 1}\right) \|(I - QQ^\top)A\|_{\text{F}}^2, \quad (9.5)$$

for any independent right sketch with $\text{rank } Y = k$. If in addition the row space of A satisfies Theorem A (for instance, it is in the conditionally Gaussian product class for the right generator, with independent or frozen right environments), then $\hat{A}_{\text{cal}}$ and $\hat{A}_{\text{fr}}$ both have exactly the two-Gaussian output law and satisfy (9.3).

Proof. Calibrated case. By Theorem 8.3, $\tilde{\Psi} = \Psi D_L^{-1}$ has $\tilde{\Psi}^\top U$ iid Gaussian, independent of the environments and of the right sketch (the weights are the exact values $w(\eta_i)$, whichever supported v is used). Moreover

$$\hat{A}_{\text{cal}} = Y(\tilde{\Psi}^\top Y)^\dagger \tilde{\Psi}^\top A = Q(\tilde{\Psi}^\top Q)^\dagger \tilde{\Psi}^\top A.$$

Frozen case. Conditionally on η_L , $\Psi^\top U = \sqrt{w(\eta_L)} \Gamma$ with Γ standard Gaussian, and a common positive scalar cancels, $(c \Psi^\top Q)^\dagger c \Psi^\top A = (\Psi^\top Q)^\dagger \Psi^\top A$; so $\hat{A}_{\text{fr}} = Q(\tilde{\Psi}^\top Q)^\dagger \tilde{\Psi}^\top A$ with $\tilde{\Psi} = \Psi / \sqrt{w(\eta_L)}$, and the conditional law of $\tilde{\Psi}^\top U$ does not depend on η_L .

Combine the two independent Gaussian ingredients. In both cases Lemma 9.1 gives (9.5). Each output is a measurable function of $\text{range } Y$ and $\tilde{\Psi}^\top U$, since $\tilde{\Psi}^\top A = \tilde{\Psi}^\top U U^\top A$ and $Q \subseteq S_L$. Under the right-side hypothesis, $\text{range } Y$ has the Gaussian law (Theorem A, Remark 8.2) and is independent of $\tilde{\Psi}^\top U$, which is standard Gaussian: this is exactly the joint law in two-sided Gaussian generalized Nyström. The bound follows from Theorem 7.2. $\square$

Corollary 9.4 (A $6r + 8$ product-query budget). *Under the hypotheses of Theorem 9.3 with the right-side condition, $k = 2r + 2$ and $p = 2k + 2$ use $k + p = 6r + 8$ product queries and give*

$$\mathbb{E}\|A - \hat{A}\|_F^2 \leq \frac{(4r+5)(2r+1)}{(2r+3)(r+1)} \tau_r^{(2)}(A) \leq 4\tau_r^{(2)}(A),$$

with no truncation step, for both $\hat{A}_{\text{cal}}$ and $\hat{A}_{\text{fr}}$. The rational prefactor is strictly below four; the last inequality is strict when $\tau_r^{(2)}(A) > 0$. The frozen-environment version needs no calibration weights at all. If $\text{rank } A \leq k$ and $p \geq \text{rank } A$, the reconstruction is exact almost surely.

Proof. $1 + \frac{k}{p-k-1} = \frac{4r+5}{2r+3} < 2$ and $1 + \frac{r}{k-r-1} = \frac{2r+1}{r+1} < 2$. $\square$

By Markov's inequality the squared-error factor is at most 20 with probability at least $4/5$. All query vectors of both constructions can be drawn before any response is observed; only the calibration weights use a response. The frozen design changes the *sampling law* (the environment is shared within the left sketch); it is not a statement about the original independently regenerated Khatri–Rao columns, for which see Proposition 9.6.

9.2 The raw estimator with independent left environments

With independent left environments and no calibration, the column scales multiply the left measurements *on the left*, and they do not cancel from least squares (Proposition 9.7 below). A bound still holds, through a weighted inverse.

Lemma 9.5 (Size-biased monotonicity). *If $w > 0$, $\mathbb{E}w = 1$, $\mathbb{E}w^2 < \infty$, and $f \geq 0$ is nonincreasing, then $\mathbb{E}[w^2 f(w)] \leq (\mathbb{E}w^2) \mathbb{E}[w f(w)]$.*

Proof. Under $d\mathbb{Q} = w d\mathbb{P}$, the increasing function w and the nonincreasing $f(w)$ have nonpositive covariance (average $(w - w')(f(w) - f(w')) \leq 0$ over two independent copies). Since $\mathbb{E}_{\mathbb{Q}}[w f(w)] = \mathbb{E}[w^2 f(w)]$, $\mathbb{E}_{\mathbb{Q}}w = \mathbb{E}w^2$ and $\mathbb{E}_{\mathbb{Q}}f(w) = \mathbb{E}[w f(w)]$, the claim follows. $\square$

Proposition 9.6 (Raw two-structured-sketch reconstruction). *Let $q = 2$ on the left, let S_L satisfy Proposition 5.3 with matrix S_0 , and use iid left probes $\psi_i = g_i \otimes h_i$. Put $\mu_2 = \mathbb{E}(h^\top S_0 h)^2 = 1 + 2 \text{tr } S_0^2 \leq 3$. For $p \geq 4(k+1)$, the raw estimator $\hat{A}_{\text{raw}} = Y(\Psi^\top Y)^\dagger \Psi^\top A$ satisfies*

$$\mathbb{E}[\|A - \hat{A}_{\text{raw}}\|_F^2 \mid \mathcal{Q}] \leq \left(1 + \mu_2 C_{\text{new}} \frac{k}{p}\right) \|(I - QQ^\top)A\|_F^2, \quad C_{\text{new}} = \frac{112}{3\pi} - \frac{4}{3}.$$

Proof. Step 1: write the weighted regression error. With the notation of Lemma 9.1, the left coordinates are $\Psi^\top Q = W^{1/2}\Xi$ and $\Psi^\top Q_\perp = W^{1/2}\Xi'$, where $W = \text{diag}(w_i)$, $w_i = h_i^\top S_0 h_i$, and Ξ, Ξ' are independent standard Gaussian and independent of W (Theorem 4.2). With $H = \Xi^\top W \Xi$, the pseudoinverse is $(W^{1/2}\Xi)^\dagger = H^{-1}\Xi^\top W^{1/2}$ and

$$A - \hat{A}_{\text{raw}} = E - QH^{-1}\Xi^\top W \Xi' F.$$

Averaging over Ξ' gives the conditional error $(1 + \mathcal{B})\|E\|_F^2$, where

$$\mathcal{B} = \mathbb{E} \text{tr}(H^{-1}\Xi^\top W^2 \Xi H^{-1}) = \mathbb{E} \sum_i w_i^2 \xi_i^\top H^{-2} \xi_i,$$

where $\xi_i^\top$ are the rows of Ξ . For $\eta > 0$ and fixed other variables, $f_i(w_i) = \xi_i^\top (H + \eta I)^{-2} \xi_i$ is nonincreasing in w_i , since

$$f'_i = -2(\xi_i^\top (H + \eta I)^{-1} \xi_i) f_i \leq 0.$$

Step 2: remove one weight by size-biased monotonicity. Since w_i is independent of the rest and $\mathbb{E}w_i = \text{tr } S_0 = 1$, Lemma 9.5 and summation give

$$\begin{aligned}\mathbb{E} \text{tr}((H + \eta I)^{-1} \Xi^\top W^2 \Xi (H + \eta I)^{-1}) &\leq \mu_2 \mathbb{E} \text{tr}(H(H + \eta I)^{-2}) \\ &\leq \mu_2 \mathbb{E} \text{tr} H^{-1}.\end{aligned}$$

Let $\eta \downarrow 0$ by monotone convergence.

Step 3: apply the weighted inverse bound. By rotation invariance of h , w has the law of $\sum_b \lambda_b(S_0) h_b^2$, which is the common-Schmidt weight law, so Theorem 7.4 (at dimension k , sample size p) gives $\mathbb{E} \text{tr} H^{-1} = p^{-1} \mathbb{E} \text{tr} S_p^{-1} \leq C_{\text{new}} k/p$. Finally $\mathbb{E}w^2 = (\text{tr } S_0)^2 + 2 \text{tr } S_0^2$. $\square$

With $\mu_2 \leq 3$, $3C_{\text{new}} < 31.651$, so $k = 2r + 2$ and $p = 32k$ give an expected squared-error factor below four, using $66r + 66$ product queries, against $6r + 8$ for the calibrated and frozen versions. The comparison reflects the constant in the weighted inverse bound more than a real gap: in simulations the coefficient $\mathcal{B}$ is about 9–11 times smaller than $\mu_2 C_{\text{new}} k/p$ (Section 12).

9.3 Two unnormalized structured sides

If the left support also satisfies the mechanism, then $U^\top \Psi = \Gamma E_0$ with a random positive diagonal E_0 , so the left measurements are multiplied on the left by E_0 . In overdetermined least squares $(E_0 M)^\dagger E_0 \neq M^\dagger$ in general.

Proposition 9.7 (Left scaling changes generalized Nyström). *Let $A = I_2$, $Y = e_1$, $\Psi^\top = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and $E_0 = \text{diag}(1, t)$, $t > 0$. The unweighted estimate $Y(\Psi^\top Y)^\dagger \Psi^\top A$ equals $e_1[1 \ 0]$, with squared error 1. With the rescaled left sketch ΨE_0 it equals*

$$e_1 \begin{bmatrix} 1 & \frac{1-t^2}{1+t^2} \end{bmatrix}, \quad \text{with squared error } 1 + \left(\frac{1-t^2}{1+t^2} \right)^2.$$

Both left sketches have the same column span.

Proof. $\Psi^\top Y = (1, 1)^\top$ has pseudoinverse $\frac{1}{2}(1, 1)$, and $\frac{1}{2}(1, 1)\Psi^\top = (1, 0)$. With E_0 : $E_0 \Psi^\top Y = (1, t)^\top$ has pseudoinverse $(1, t)/(1 + t^2)$, and $(1, t)E_0 \Psi^\top / (1 + t^2) = (1 + t^2, 1 - t^2)/(1 + t^2)$. The error matrix is $\begin{bmatrix} 0 & -c \\ 0 & 1 \end{bmatrix}$ with $c = (1 - t^2)/(1 + t^2)$. $\square$

So column-span equivalence alone cannot give the two-sided estimator a Gaussian law; the example refutes that invariance, not the existence of good structured algorithms. The remedies are exactly the constructions of Theorem 9.3: rescale by the known weights (calibration, equivalently $\Psi \mapsto \Psi E_0^{-1}$), or make the left scale common to all columns (frozen environment), which then cancels.

Remark 9.8 (Relation to an open question on general inputs). Camaño, Epperly, Meyer and Tropp [4, Remark 2.9] analyse generalized Nyström with structured sketches through oblivious subspace injections, with $k = \gamma r$ right and $p = \gamma^2 r$ left columns for an oversampling factor γ , and ask whether $p = O(\gamma r)$ suffices without truncation. On the class studied here both stages are Gaussian-equivalent, $\gamma = O(1)$, and the question does not arise. It concerns general inputs, where neither the support structure nor the left calibration is available. One concrete bridge would be a bound $\mathbb{E}_\Psi \|(\Psi^\top Q)^\dagger \Psi^\top E\|_F^2 \leq C \|E\|_F^2$ for $Q^\top E = 0$ and economical structured Ψ ; Lemma 9.1 is its Gaussian case.

10 Calibrated trace estimation

The estimator combines a Nyström approximation with a calibrated Hutchinson correction, as in Nyström++ [13] and Hutch++ [7]. We first prove the Gaussian Nyström residual bound it needs, by a drift argument; the same argument with a product-dependent constant governs unrestricted inputs in “Product-Probe Drift, Kronecker Nyström Approximation, and Trace Estimation” [14].

Lemma 10.1 (Residual update). *Let $A \succeq 0$, let C_t be the Nyström approximation from test vectors $z_1, \dots, z_t$ and $R_t = A - C_t$. If $z_{t+1}^\top R_t z_{t+1} > 0$ then $R_{t+1} = R_t - R_t z_{t+1} z_{t+1}^\top R_t / (z_{t+1}^\top R_t z_{t+1})$; otherwise $R_{t+1} = R_t$. In particular $0 \preceq R_{t+1} \preceq R_t$.*

Proof. By (3.2), $R_t = A^{1/2}(I - P_t)A^{1/2}$ with P_t the projector onto $\text{range}(A^{1/2}Z_t)$. Adding $z = z_{t+1}$ adds to P_t the rank-one projector onto $x = (I - P_t)A^{1/2}z$ when $x \neq 0$, and $A^{1/2}xx^\top A^{1/2}/\|x\|^2 = R_t z z^\top R_t / (z^\top R_t z)$. $\square$

Lemma 10.2 (Gaussian drift). *Let $R \succeq 0$, $R \neq 0$, and $g \sim N(0, I)$. Then*

$$\mathbb{E} \frac{g^\top R^2 g}{g^\top R g} \geq \frac{\text{tr } R^2}{\kappa_G \text{tr } R}, \quad \kappa_G = \exp(2 - \gamma - \log 2) = \frac{e^{2-\gamma}}{2} = 2.0743278 \dots$$

Proof. Step 1: tilt by the numerator. Let $X = g^\top R g$, $Y = g^\top R^2 g$, and $d\mathbb{Q} = (Y/\text{tr } R^2) d\mathbb{P}$. Jensen’s inequality gives

$$\mathbb{E}[Y/X] = \text{tr } R^2 \mathbb{E}_{\mathbb{Q}}[X^{-1}] \geq \text{tr } R^2 \exp(-\mathbb{E}_{\mathbb{Q}} \log X).$$

For $\beta \in (0, 1)$, a second use of Jensen yields

$$\mathbb{E}_{\mathbb{Q}} \log X \leq \beta^{-1} \log \mathbb{E}_{\mathbb{Q}} X^\beta.$$

The Gaussian gamma integrals below also ensure the required logarithmic integrability.

Step 2: bound the tilted fractional moment. Diagonalize $R = \sum_i \mu_i u_i u_i^\top$ and put $z_i = (u_i^\top g)^2$, iid χ_1^2 with $\mathbb{E} z_i^p = c(p) := 2^p \Gamma(p + \frac{1}{2}) / \sqrt{\pi}$. By Hölder with exponents $1 + \beta$ and $(1 + \beta)/\beta$, and Minkowski in $L^{1+\beta}$,

$$\begin{aligned} \mathbb{E}[z_i X^\beta] &\leq c(1 + \beta)^{\frac{1}{1+\beta}} (\mathbb{E} X^{1+\beta})^{\frac{\beta}{1+\beta}} \\ &\leq c(1 + \beta)^{\frac{1}{1+\beta}} (c(1 + \beta)^{\frac{1}{1+\beta}} \text{tr } R)^\beta \\ &= c(1 + \beta) (\text{tr } R)^\beta. \end{aligned}$$

Consequently,

$$\mathbb{E}_{\mathbb{Q}} X^\beta = \frac{\sum_i \mu_i^2 \mathbb{E}[z_i X^\beta]}{\text{tr } R^2} \leq c(1 + \beta) (\text{tr } R)^\beta.$$

Step 3: let the moment order tend to one. Substituting in Step 1 gives

$$\mathbb{E}[Y/X] \geq \frac{\text{tr } R^2}{c(1 + \beta)^{1/\beta} \text{tr } R}.$$

As $\beta \downarrow 0$,

$$c(1 + \beta)^{1/\beta} \longrightarrow \exp\left(\frac{d}{dp} \log c(p) \Big|_{p=1}\right) = \exp(\log 2 + \psi(\tfrac{3}{2})) = \kappa_G,$$

where $\psi(\frac{3}{2}) = 2 - \gamma - 2 \log 2$. $\square$

Lemma 10.3 (Nyström residual). *If $A \succeq 0$ has range S satisfying Theorem A, and C is the Nyström approximation from m probes, then $\mathbb{E}\|A - C\|_F^2 \leq \kappa_G(\text{tr } A)^2/m$.*

Proof. $\|A - C\|_F^2$ is a function of C , which is range-determined, so we may use Gaussian probes. The case $A = 0$ is immediate, so put $T = \text{tr } A > 0$. By Lemmas 10.1 and 10.2 (applied to $V^\top R_t V$ and $V^\top z_{t+1} \sim N(0, I_R)$, independent of R_t),

$$\mathbb{E}[\text{tr } R_t - \text{tr } R_{t+1} \mid R_t] \geq \frac{\text{tr } R_t^2}{\kappa_G \text{tr } R_t},$$

with $0/0 = 0$. The trace loss telescopes, so

$$\sum_{t < m} \mathbb{E} \frac{\text{tr } R_t^2}{\text{tr } R_t} \leq \kappa_G T.$$

Since $\text{tr } R_t \leq T$ and $\text{tr } R_t^2 \geq \text{tr } R_m^2$ (as $0 \preceq R_m \preceq R_t$), each term is at least $\mathbb{E} \text{tr } R_m^2 / T$. Thus $m \mathbb{E}\|R_m\|_F^2 / T \leq \kappa_G T$, as required. $\square$

The estimator. Let $A \succeq 0$ have range S in the conditionally Gaussian product class. Given $\varepsilon \in (0, 1)$, put $m = s = \lceil \sqrt{6\kappa_G}/\varepsilon \rceil$.

- If $2m \geq n_1$: draw a random good environment h , query $A(e_i \otimes h)$ for $i \leq n_1$, read w from a nonzero response, and return $\hat{T} = \text{tr}(L_h^\top A L_h)/w$ (return 0 if all responses vanish).
- Otherwise: draw m independent Gaussian product probes Z , form $Y = AZ$ and $C = Y(Z^\top Y)^\dagger Y^\top$, take a nonzero column v of Y (if none, return 0), draw s independent test probes $\omega_j = L_{\eta_j} g_j$, compute $w_j = \|L_{\eta_j}^\top v\|^2 / \|v\|^2$, and return

$$\hat{T} = \text{tr } C + \frac{1}{s} \sum_{j=1}^s \frac{\omega_j^\top (A \omega_j - C \omega_j)}{w_j}. \quad (10.1)$$

All query vectors are chosen before any response is seen, and every query is a legal product; $C \omega_j$ is computed by arithmetic.

Proof of Theorem E. The first branch uses n_1 queries and is exact by Theorem C(d). In the second branch, $2m < n_1$ queries are used. Let $\tilde{R} = V^\top (A - C)V$ and $g_j = V^\top \omega_j / \sqrt{w_j}$. The range of $A - C$ lies in S , so $\omega_j^\top (A - C) \omega_j / w_j = g_j^\top \tilde{R} g_j$. By Theorem 8.3, the g_j are iid $N(0, I_R)$ and independent of Z and hence of C . Therefore $\mathbb{E}[\hat{T} \mid C] = \text{tr } C + \text{tr } \tilde{R} = T$ and $\text{Var}(\hat{T} \mid C) = 2\|A - C\|_F^2/s$. By Lemma 10.3,

$$\text{Var } \hat{T} = \mathbb{E} \text{Var}(\hat{T} \mid C) \leq \frac{2\kappa_G}{ms} T^2,$$

and Chebyshev's inequality with $ms \geq 6\kappa_G/\varepsilon^2$ gives failure probability at most $1/3$. If $m \geq R$, the lines $[V^\top z_i]$ are iid uniform and span $\mathbb{R}^R$ almost surely, so $C = A$ and $\hat{T} = T$. $\square$

The leading constant is $2\sqrt{6\kappa_G} = 7.0557684$, so the count is $\min\{7.06/\varepsilon + 2, n_1\}$ up to rounding. A median of $O(\log(1/\delta))$ independent repetitions gives confidence $1 - \delta$ (the median need not be unbiased), and all queries can still be fixed in advance.

Remark 10.4 (What the theorem is). After calibration, the approximation and test probes project to iid Gaussian vectors on S , so the second branch has exactly the law of Gaussian Nyström++

on the R -dimensional support, equivalently on the compressed matrix B of Theorem C. The $O(1/\varepsilon)$ rate is therefore the familiar Gaussian rate [7, 13], realized with product queries, and not a new trace-estimation principle. It is informative only when n_1 (and rank A) is much larger than about $7/\varepsilon$; otherwise n_1 product queries give the trace exactly. The cap uses n_1 , which is known in advance, rather than the unknown rank A . For general PSD inputs, product-query trace estimation is a different and harder problem; see the product-probe companion [14] and the general-input lower bounds of [11].

11 Where the equivalence stops

11.1 A Gaussian range law is not a Gaussian embedding law

Take $u = e_1 \otimes f$ with f a unit vector, and the Gaussian Khatri–Rao probe $\omega = g \otimes h$. The projected coordinate is $\langle u, \omega \rangle = g_1 \langle f, h \rangle$, a product of two independent standard Gaussians: its second moment is 1 and its fourth moment is 9, against 3 for a Gaussian. One nonzero probe already spans this one-dimensional support, so every range-determined output is exact and has the Gaussian law, while the sketch Gram law is not Gaussian. In general, under (4.2),

$$(V^\top \Omega)(V^\top \Omega)^\top = GD^2G^\top,$$

and D cannot be discarded. Embedding distortion, inverse-Gram moments and numerical conditioning are different observables. Nothing here supersedes Khatri–Rao embedding results [2, 16, 1] or oblivious subspace injection bounds [4].

11.2 A hypothesis on the leading eigenvectors alone is insufficient

Let $n_1 = n_2 = 2$, $u_1 = e_1 \otimes e_1$, $u_2 = e_2 \otimes e_2$, and $A = \mu_1 u_1 u_1^\top + \mu_2 u_2 u_2^\top$ with $\mu_1 \geq \mu_2 > 0$. Each u_a separately has the common-Schmidt form, and the leading eigenvector u_1 alone meets the mechanism. The support $S = \text{span}\{u_1, u_2\}$ does not: the projected probe $(g_1 h_1, g_2 h_2)$ has conditional covariance $\text{diag}(h_1^2, h_2^2)$, and its line is not uniform. The effect on the output is explicit.

Proposition 11.1 (Head-only counterexample). *Let C be the one-column Nyström approximation of A and $y = \sqrt{\mu_2/\mu_1} \in (0, 1]$. For a dense Gaussian probe $\mathbb{E} \text{tr}(A - C) = \sqrt{\mu_1 \mu_2}$. For the Gaussian Khatri–Rao probe*

$$\mathbb{E} \text{tr}(A - C) = \sqrt{\mu_1 \mu_2} \Phi(y), \quad \Phi(y) = \frac{2y + \frac{2}{\pi}(1 - y^2) \log(1/y)}{1 + y^2}.$$

$\Phi(1) = 1$, $\Phi(y) > 1$ for $0 < y < 1$, and $\Phi(y) \sim \frac{2}{\pi} \log(1/y)$ as $y \rightarrow 0$. Even at $y = 1$, where the trace error is the constant μ_1 for both probes, the output laws differ: $\text{range } C$ is the line of slope c spanned by the projected probe, and $\log |c|$ has variance $\pi^2/2$, against $\pi^2/4$ for the Gaussian.

Proof. Step 1: express the error through the projected slope. For a probe with projected coordinates (c_1, c_2) and $c = c_2/c_1$,

$$\text{tr}(A - C) = \frac{\mu_1 \mu_2 (c_1^2 + c_2^2)}{\mu_1 c_1^2 + \mu_2 c_2^2} = \frac{\mu_1 \mu_2 (1 + c^2)}{\mu_1 + \mu_2 c^2}.$$

Normalize $\mu_1 = 1$, $\mu_2 = y^2$. The ratio of the mean error to $\sqrt{\mu_1\mu_2} = y$ is $\mathbb{E} y(1 + c^2)/(1 + y^2c^2)$. The decomposition

$$\frac{1 + c^2}{1 + y^2c^2} = y^{-2} + \frac{1 - y^{-2}}{1 + y^2c^2}$$

reduces the calculation to one Cauchy integral.

Step 2: average the random Cauchy scale. If c is Cauchy with scale a , convolution of Cauchy kernels gives $\mathbb{E}[1/(1 + y^2c^2)] = 1/(1 + ay)$. For a Gaussian probe, c is standard Cauchy ($a = 1$), and the ratio equals $y[y^{-2} + (1 - y^{-2})/(1 + y)] = 1$. For a Khatri–Rao probe, $c = (g_2/g_1)(h_2/h_1)$ is Cauchy with random scale $a = |h_2/h_1|$, whose density is $2/(\pi(1 + a^2))$ on $a > 0$, and a partial-fraction computation gives

$$\mathbb{E}_a \frac{1}{1 + ay} = \frac{2}{\pi} \int_0^\infty \frac{da}{(1 + a^2)(1 + ay)} = \frac{1 + \frac{2}{\pi}y \log y}{1 + y^2}.$$

Substitution yields the stated expression for Φ .

Step 3: show the gap and distinguish the equal-eigenvalue laws. We have

$$\Phi(y) - 1 = \frac{(1 - y)[\frac{2}{\pi}(1 + y) \log(1/y) - (1 - y)]}{1 + y^2}.$$

The inequality $\log(1/y) \geq 2(1 - y)/(1 + y)$ gives $\frac{2}{\pi}(1 + y) \log(1/y) \geq \frac{4}{\pi}(1 - y) > 1 - y$ for $y < 1$. Finally $\log |c|$ is a sum of two independent copies of $\log |g_2/g_1|$, each of variance $2 \cdot \pi^2/8 = \pi^2/4$, against one copy for the Gaussian. $\square$

The requirement is on the *entire* relevant support, spectral tail included. A certificate for the leading r eigenvectors says nothing about the joint law of the head and tail coordinates on which the error bounds rest.

11.3 Weight-sensitive surrogates do not cancel

Deterministic bounds are often proved with the candidate coefficient matrix $(G_1D)^\dagger$, which produces the surrogate $G_2D(G_1D)^\dagger$. The weights do not cancel from it. The range also admits the coefficient $D^{-1}G_1^\dagger$, which produces $G_2G_1^\dagger$, and the second is never worse.

Proposition 11.2 (Surrogate domination). *Let $G_1 \in \mathbb{R}^{r \times k}$ have full row rank, D be invertible diagonal, and G_2 be an independent standard Gaussian matrix. For every $\Lambda_2 \succeq 0$,*

$$\begin{aligned} \mathbb{E}_{G_2} \|\Lambda_2^{1/2} G_2 D (G_1 D)^\dagger\|_F^2 &= \text{tr } \Lambda_2 \|D (G_1 D)^\dagger\|_F^2 \\ &\geq \text{tr } \Lambda_2 \|G_1^\dagger\|_F^2 \\ &= \mathbb{E}_{G_2} \|\Lambda_2^{1/2} G_2 G_1^\dagger\|_F^2. \end{aligned}$$

Proof. The equalities are Gaussian second moments. $D(G_1D)^\dagger$ is a right inverse of G_1 , since $G_1D(G_1D)^\dagger = I_r$, and $G_1^\dagger$ is the right inverse of minimal Frobenius norm. $\square$

This is the reason the weighted inverse route in Theorem 7.4 yields a coefficient about $31.65 r/k$ whereas the exact range law yields $r/(k - r - 1)$. There is no conflict: the first is one valid candidate bound, while the approximation depends on the whole range.

11.4 Two unnormalized structured sides; exact arithmetic

Proposition 9.7 shows that independent left scales change generalized Nyström even when the left span is unchanged, so the two-sided estimator with independently regenerated structured sketches on both sides has no automatic Gaussian law; Proposition 9.6 bounds it with a worse constant. Finally, all statements concern exact projectors and pseudoinverses. Very small column scales can make a floating-point implementation discard a column or misjudge a rank; column equilibration may help but needs its own numerical analysis.

12 Numerical illustrations

All experiments below are numerical checks of proved statements, run in Julia 1.13 in double precision with fixed seeds. They are diagnostics, not proofs. “Common-Schmidt” supports are generated from random orthonormal $e_{a,b}$ and f_b as in (5.1). The six Julia scripts and their rerun outputs are included in the [repository checks directory](#). An independent Python audit also checks the added weighted-inverse constant over 10,000 admissible budgets, the radial distribution-function bound, the closed-form counterexample, and the matrix identities. These finite checks supplement the proofs and do not certify floating-point stability.

Exact laws. Table 1 compares Gaussian Khatri–Rao Nyström with dense Gaussian Nyström on a common-Schmidt input ($n_1 = 12$, $n_2 = 3$, $\ell = 2$, weights $(0.7, 0.3)$, $R = 6$, eigenvalues $10, 5, 1, 0.5, 0.3, 0.2$, $r = 2$, $k = 4$, 20,000 trials). Means and quantiles agree within Monte Carlo error, and both sit well below the bound $(1 + r/(k - r - 1))\tau_r = 6$ of Corollary 7.5.

	mean $\text{tr}(A - C)$	mean $\text{tr}(A - C_r)$	10%	50%	90%
Khatri–Rao	1.7007	2.8526	2.1246	2.5284	3.9731
Gaussian	1.6933	2.8431	2.1241	2.5303	3.9734

Table 1: Common-Schmidt Khatri–Rao versus Gaussian Nyström; quantiles are of $\text{tr}(A - C_r)$.

Further law checks:

- *Hurwitz pair* (Example 5.4): the identities hold to 1.6×10^{-16} ; the angle of $V^\top \omega$ in eight bins of width $\pi/8$ on $[0, \pi)$ has frequencies 0.124–0.126 (uniform: 0.125), 2×10^5 samples.
- *Spherical factors* (Corollary 4.4; $R = 3$): the direction of $V^\top \omega$ has $\|\mathbb{E} u u^\top - I/3\| = 0.0020$ and $\mathbb{E} u_1^4 = 0.1990$ against 0.2 for the uniform law on S^2 .
- *Single-core tensor train* (Proposition 5.6; $q = 3$, $\chi = 2$): $\mathbb{E} \|\omega\|^2/N = 1.002$, and the direction histogram is uniform within 0.003. The fourth moment of one projected coordinate over its squared variance is 13.4, against 3 for a Gaussian: the range law is Gaussian, the embedding law is not.
- *Head-only counterexample* (Proposition 11.1): Table 2; 2×10^6 samples per row. The variance of the log absolute slope is 4.9255 for Khatri–Rao ($\pi^2/2 = 4.9348$) and 2.4674 for Gaussian ($\pi^2/4 = 2.4674$). A quadrature of Φ against the product-of-Cauchy density agrees with the closed form to six digits.

(μ_1, μ_2)	Gaussian MC	$\sqrt{\mu_1 \mu_2}$	Khatri–Rao MC	$\sqrt{\mu_1 \mu_2} \Phi(y)$	$\Phi(y)$
(1, 0.25)	0.4997	0.5000	0.5327	0.5324	1.065
(1, 0.01)	0.1000	0.1000	0.1634	0.1635	1.635
(1, 10^{-4})	0.0101	0.0100	0.0295	0.0295	2.951

Table 2: One-column Nyström trace error in the head-only counterexample.

Dimension counts. For the Segre cone with $(n_j) = (2, 2), (2, 2, 2), (2, 2, 2, 2, 2), (3, 4)$ and $(3, 3, 3)$, the generic Jacobian rank of $(x_j) \mapsto x_1 \otimes \cdots \otimes x_q$ equals $\delta + 1$ in every case. The generic rank of the differential of the map from k product probes to $\text{range}(V^\top \Omega) \in \text{Gr}(k, R)$, for a random R -dimensional V , is $\min\{k\delta, k(R - k)\}$ in all tested cases (Table 3), so the bound $R \leq k + \delta$ of Theorem B(ii) is exactly the point where the image stops being full-dimensional. The hook probe of Example 6.1 with $(n_j) = (3, 2, 4)$ reproduces $V^\top \omega = (z_0, z_1, \dots, z_q)$ to 1.9×10^{-16} on a support of dimension $\delta + 1 = 7$.

(n_j)	k	R	image dim	$k\delta$	$k(R - k)$	$R \leq k + \delta$
(2, 2, 2)	1	4	3	3	3	yes
(2, 2, 2)	1	5	3	3	4	no
(2, 2, 2)	2	5	6	6	6	yes
(2, 2, 2)	2	6	6	6	8	no
(2, 2, 2)	3	7	9	9	12	no
(2, 2, 2)	4	8	12	12	16	no
(3, 3)	2	6	8	8	8	yes
(3, 3)	2	7	8	8	10	no

Table 3: Generic dimension of the image of product-probe ranges in $\text{Gr}(k, R)$.

The n_1 -dimensional reduction. On a two-factor common-Schmidt input ($n_1 = 12$, $n_2 = 4$, weights $(0.6, 0.3, 0.1)$, $R = 4$, eigenvalues $10, 3, 1, 0.2$) with a random frozen environment: $\|V^\top L L^\top V - wI\| = 4.9 \times 10^{-16}$; the one-query product $A((L^\top v/w) \otimes h)$ matches Av to relative error 1.6×10^{-16} , and w read from a response matches to 1.5×10^{-16} ; for an unsupported v the error is 2.95, as it must be. The compressed matrix has spectrum $(10, 3, 1, 0.2, 0, \dots)$; Nyström from the $n_1 = 12$ queries $e_i \otimes h$ recovers A to 4.7×10^{-16} with $\text{tr } B = \text{tr } A$; five powers computed with one query each match to 7.5×10^{-16} . At third order ($n = (6, 2, 2)$, entangled f_b), over 1000 random frozen product environments the worst relative error of the one-query identity is 4.8×10^{-15} . For a rectangular input, four cycles of $AA^\top$ with two queries each match to 4.1×10^{-16} ; the compressed matrix has singular values $(5, 3, 2, 1, 0.5, 0, \dots)$; exact recovery from n_1 right and n_1 left product queries has error 5.4×10^{-16} ; the log-determinant identity of Corollary 8.4 holds to rounding, and a lifted eigenvector has residual 3.2×10^{-15} .

Generalized Nyström. Table 4 uses a common-Schmidt input with $n_1 = 16$, $n_2 = 3$, rank 8, singular values $1, 0.8, 0.5, 0.3, 0.2, 0.15, 0.1, 0.05$, a Khatri–Rao right sketch with $k = 4$, and $p = 10$ left columns (6000 trials). The calibrated and frozen versions match the Gaussian law and the prediction of (9.5); the raw independent-column version is visibly worse, consistent with Proposition 9.7. In a second test with both environments frozen (left and right supports with different Schmidt weights, rank 8), the mean and median squared errors were 2.921 and 2.245

against 2.892 and 2.246 for Gaussian sketches. The left-scaling counterexample is reproduced exactly (error 1.36 at $t = 0.5$ and $t = 2$).

estimator	mean $\ A - \hat{A}\ _F^2$	s.e.	median
prediction $(1 + \frac{k}{p-k-1})\mathbb{E}\ (I - QQ^\top)A\ _F^2$	0.5531	0.0034	–
Gaussian both sides	0.5589	0.0052	0.4562
calibrated (9.4)	0.5477	0.0045	0.4529
frozen left environment	0.5539	0.0050	0.4539
raw, independent left environments	0.6918	0.0070	0.5456

Table 4: Generalized Nyström with structured sketches.

For the raw estimator, the exact coefficient $\mathcal{B}$ of Proposition 9.6 was estimated for S_0 of rank one, $I_2/2$ and $I_4/4$ and $(k, p) \in \{(4, 20), (4, 40), (8, 36), (8, 72)\}$; the bound $\mu_2 C_{\text{new}} k/p$ exceeded it by factors 8.7 to 10.7. The $6r + 8$ budget of Corollary 9.4 has worst-case factor $(4r + 5)(2r + 1)/((2r + 3)(r + 1))$, which is 2.70, 3.10, 3.53, 3.74, 3.97 for $r = 1, 2, 5, 10, 100$.

Trace estimation. Table 5 uses a common-Schmidt input with $n_1 = 200$, $n_2 = 3$, weights $(0.5, 0.3, 0.2)$, $R = 60$, eigenvalues $1/j$, and 2000 trials per row. The estimator is unbiased within two standard errors, and the variance is about 13 to 16 times below the bound $2\kappa_G/(ms)$. With $m = R = 60$ the maximum relative error over 20 rerun trials was 3.2×10^{-12} , illustrating exactness once $m \geq \text{rank } A$.

$m = s$	mean/ T	s.e.	Var/ T^2	bound $2\kappa_G/(ms)$
5	0.9957	0.0024	1.12×10^{-2}	1.66×10^{-1}
10	1.0010	0.0013	3.23×10^{-3}	4.15×10^{-2}
20	1.0005	0.0006	6.56×10^{-4}	1.04×10^{-2}

Table 5: Calibrated product-query trace estimator.

13 Open problems

1. *Stability.* Extend the exact identification to supports whose conditional covariance is only approximately scalar, with explicit control of head–tail correlations and inverse conditioning. This is the step that would turn the explanatory class into an algorithmic one for inputs of rank larger than n_1 (“structured signal plus unrestricted tail”).
2. *Tensor trains with growing rank.* Do Gaussian tensor-train probes admit supports of growing dimension, beyond the single-core family, on which Theorem A holds? The dimension bound $R \leq \dim \mathcal{T}_\chi$ (Remark 6.3) permits it; no construction or obstruction beyond the stated families is proved here.
3. *Sharpness of the fixed- k ceiling.* For independent-factor product laws, is $R \leq k + \delta$ attained by range-law equality at a single sketch size k , without equality at $k = 1$?
4. *Economical structured left sketches on general inputs.* Prove $\mathbb{E}_\Psi \|(\Psi^\top Q)^\dagger \Psi^\top E\|_F^2 \leq C \|E\|_F^2$ for $Q^\top E = 0$ and a structured Ψ with $p = O(k)$ columns, without the support hypothesis. This would engage the question of [4, Remark 2.9].

5. *Classification*. For two independent Gaussian factors the conditional mechanism is characterized by Proposition 5.3 and its mirror. Characterize all supports satisfying projective uniformity (Theorem A(iii)) for Gaussian Khatri–Rao probes, without conditioning on a factor.
6. *Finite precision*. Quantify how small environment scales w affect the one-query identity and the calibrated estimators in floating point, and whether a best-of-a-few environment choice suffices.

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